

FieldCommand Incident Management System

Open-Source · Offline-First · Field-Deployable

# FIELDCOMMAND

Incident Management System v1.0

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## INSTALLATION GUIDE

ASUS RT-BE58 Go · UniFi Switch Lite 16 PoE · Raspberry Pi 5 · Pironman MAX 5

ICS/NIMS All-Hazards · Amateur Radio EMCOMM · August 07, 2026

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## 1. Overview & What's Included

FieldCommand is a complete off-grid emergency communications and ICS management system built on two Raspberry Pi 5 units — one running all 49 web-based tools covering ICS forms, net logging, resource tracking, FEMA cost documentation, APRS, Winlink, and personnel accountability; one dedicated to the AMPRNet / 44Net gateway. WAN connectivity is optional and configurable — your group chooses the primary source (cellular modem, satellite dish, site Ethernet, or tethered phone) and whether to add a secondary source for automatic failover, all managed by the ASUS RT-BE58 Go Wi-Fi 7 router. Two additional ASUS RT-BE58 Go routers extend EMCOMM-NET as AiMesh nodes for large venues. Any phone, tablet, or laptop connects to EMCOMM-NET and reaches all tools at <http://192.168.50.1> — no internet, no app, no per-device setup required. When WAN is available, NWS alerts, APRS-IS, and HF propagation data activate automatically.

**SAR Network Operations:** The Starcom Net Logger includes a dedicated SAR Net mode (Search & Rescue) accessible directly from the Starcom / Public Safety dashboard. It provides a SAR-specific check-in log with unit type tracking, zone assignments, and resource status. The Starcom Resource Map provides the tactical overlay for SAR unit positions. These features are integrated into the main FieldCommand platform — no separate module is required.

| COMPONENT           | DESCRIPTION                                                     | COMPONENT            | DESCRIPTION                                                |
|---------------------|-----------------------------------------------------------------|----------------------|------------------------------------------------------------|
| 30 Web Pages        | Full dashboard, net logs, ICS forms, maps, roster, cheat sheets | Dead Man's Switch    | Per-net inactivity monitor with warning and trigger states |
| Net Control Logger  | Amateur + Starcom nets, FCC autofill, ICS-309 export            | Pre-Flight Checklist | GO/CAUTION/NO-GO deployment readiness check                |
| ICS Platform        | Command, Operations, Planning, Logistics, Finance sections      | HF Propagation       | Solar indices, band conditions, A/K-index                  |
| APRS Tactical Map   | Direwolf + YAAC merged, offline tiles, overlays                 | Repeater Database    | RepeaterBook CSV, filter by band, mode, ARES affiliation   |
| Member Roster       | your organization directory with certs, equipment, activations  | Reference Library    | Upload and serve field reference docs across EMCOMM-NET    |
| NTS Radiogram       | ARRL-formatted radiogram generator with traffic log             | Kiwix Library        | WikiMed, Wikipedia, iFixit — offline at port 8081          |
| ICS-213/214/309     | Fillable ICS forms with print output and incident archiving     | FCC Callsign Lookup  | ~800K licensees — instant offline search                   |
| Winlink Form Import | Import Winlink Express XML forms to incident archive            | Pat Winlink          | Browser-based backup Winlink client at port 8090           |
| JS8Call Integration | Dashboard card opens JS8Call web API on Windows laptop          | ICS Planning P       | Interactive 15-phase planning cycle guide                  |

# 1. Hardware Requirements

## Supported Hardware — Raspberry Pi Servers

FieldCommand v1.0 uses two Raspberry Pi 5 units: one as the FieldCommand application server and one as the dedicated AMPRNet / 44Net gateway. Keeping them separate means the gateway can be rebooted, reconfigured, or upgraded without affecting the FieldCommand server, net logs, or active incidents.

| UNIT                | HARDWARE                                                                                                                 | STORAGE                                                         | ROLE                                                                                                                                                                                                               |
|---------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FieldCommand Server | Raspberry Pi 5 — 16 GB RAM / Pironman MAX 5 tower enclosure (dual M.2 NVMe, OLED, active cooling)                        | 2× 1 TB NVMe SSD / configured as RAID 1                         | Runs all FieldCommand services: web server, FCC database, net logging, ICS platform, APRS, Kiwix, Winlink, maps, and health monitor. Static IP: 192.168.50.1                                                       |
| 44Net Gateway       | Raspberry Pi 5 — 16 GB RAM / Argon NEO 5 M.2 BRED case (compact, passive-cooled, M.2 SATA slot) / Pi OS Desktop (64-bit) | 256 GB M.2 SATA SSD (OS + WireGuard + routing + status service) | Dedicated AMPRNet / 44Net gateway. Runs WireGuard tunnel to amprgw.ampr.org, ip routing between EMCOMM-NET and 44.0.0.0/8, and a status page at 192.168.50.2. Completely independent from the FieldCommand server. |

## Network Hardware

Wi-Fi is provided by the ASUS RT-BE58 Go travel router — neither Pi runs a Wi-Fi hotspot. The UniFi Switch Lite 16 PoE provides 16 wired gigabit ports for both Pi servers, the Windows laptop, four operator workstations, a network printer, and room for expansion.

| DEVICE                   | ROLE                                                                    | NOTES                                                                                                                                                                                                                                                                                                   |
|--------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASUS RT-BE58 Go          | Wi-Fi 7 access point / DHCP server / WAN gateway                        | Dual-band Wi-Fi 7 (802.11be). 2.5G LAN uplink to switch. USB-C 18W power. Handles EMCOMM-NET Wi-Fi, DHCP for 192.168.50.100-200, and optional WAN routing.                                                                                                                                              |
| UniFi Switch Lite 16 PoE | 16-port gigabit managed switch / 8x PoE + 8x non-PoE / 2x 1G SFP uplink | Port layout: 1=ASUS router uplink, 2=FieldCommand Pi, 3=44Net Gateway Pi, 4=Windows laptop, 5=color MFP printer, 6-9=Pi 500 workstations, 10=Satellite dish (optional), 11-12=AiMesh nodes, 13-16=spare. ~\$200. PoE ports power cameras, future UniFi APs, or PoE devices without a separate injector. |

## Accessories & Cables

| ITEM                                                                    | REQUIRED?   | PURPOSE                                                                                                                                |
|-------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------|
| USB-A drive (32 GB+, labeled FIELDCOMMAND)                              | Recommended | Auto-backup trigger — insert to back up all runtime data instantly                                                                     |
| External USB hard drive 1 TB+ (e.g. LaCie Rugged or LaCie Mobile Drive) | Recommended | Incident archive and full system backup. LaCie Rugged is shock/drop/crush resistant. Label FIELDCOMMAND for auto-backup compatibility. |
| USB GPS receiver (u-blox or GlobalSat)                                  | Optional    | Live position for tactical map and NWS alerts                                                                                          |
| Powered USB hub (4- or 7-port, with power supply)                       | Optional    | Recommended if connecting GPS + backup drive + TNC simultaneously. Must be powered — unpowered hubs can cause instability.             |
| USB Printer (e.g. Brother HL-L2350DW, HP LaserJet)                      | Optional    | Shared across EMCOMM-NET via CUPS (installed automatically). Or connect a Wi-Fi printer directly to EMCOMM-NET.                        |

## Amateur Radio Accessories (amateur radio group only)

The items in this section are only needed if a licensed amateur radio group is involved in the deployment. If FieldCommand is being deployed by a public safety agency or served organization without amateur radio operators, skip this section entirely.

| ITEM                                             | REQUIRED?     | PURPOSE                                                                                                                                                                                                                |
|--------------------------------------------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| USB TNC — Digirig Mobile                         | If using APRS | USB soundcard + serial CAT interface in one unit. Connects to any radio with a 3.5mm audio port. Enables APRS transmit/receive via Direwolf and YAAC. FieldCommand assigns stable /dev/tnc0 device name via udev rule. |
| USB TNC — Signalink USB (alternative to Digirig) | If using APRS | Tigertronics Signalink USB — USB audio interface, requires rig-specific audio cable. Widely used and well-supported. Same udev /dev/tnc0 assignment as Digirig.                                                        |
| Mobilinkd TNC4 (Bluetooth alternative)           | If using APRS | Bluetooth APRS TNC — pairs with YAAC on the Pi or with a phone running APRSdroid. Useful when a USB port is not available or for portable deployments.                                                                 |
| Rig interface cables                             | If using TNC  | Radio-specific audio + PTT cable connecting the TNC to the transceiver. Digirig: cables available at digirig.net by radio model. Signalink: requires separate Tigertronics cable for each rig.                         |
| Windows laptop (Winlink Express + JS8Call)       | If using HF   | Any Windows 10/11 laptop with USB-A port and Wi-Fi. Required for Winlink Express, VARA HF, and JS8Call. Connects to the IC-7300 via USB-B cable. See Part 2E of this guide for setup.                                  |

## USB Hub & Multi-Device Setup

The Pi 5 has four USB ports. A powered USB hub lets you connect GPS, TNC, backup drive, and other peripherals simultaneously. FieldCommand uses udev rules to assign stable device names regardless of hub port or plug order. The TNC entries in the table below apply only to amateur radio group deployments — if no TNC is used, the GPS and backup drive rows still apply.

| DEVICE                                          | STABLE NAME                 | HOW ASSIGNED                                                                                                               | SERVICE                      |
|-------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------|
| USB GPS receiver                                | /dev/gps0                   | udev rule matches by USB vendor/product ID                                                                                 | gpsd → tactical map          |
| Digirig Mobile TNC                              | /dev/tnc0<br>/dev/digirig   | udev rule matches by vendor ID + product string "Digirig Mobile". Distinguished from GPS even though both use CP2102 chip. | YAAC / Direwolf APRS         |
| Signalink USB TNC                               | /dev/tnc0<br>/dev/signalink | udev rule matches by Texas Instruments PCM2904 chip. No conflict with GPS.                                                 | YAAC / Direwolf APRS         |
| LaCie / USB backup drive (labeled FIELDCOMMAND) | /dev/sdX<br>(auto-mounted)  | udev backup rule triggers on label FIELDCOMMAND. Drive letter assigned dynamically.                                        | fieldcommand-backup@.service |

Both the Digirig Mobile TNC and GlobalSat BU-353-S4 GPS use the same USB chip (Silicon Labs CP2102, vendor 10c4:ea60). FieldCommand distinguishes them by USB product description string. The GPS rule excludes Digirig; the TNC rule requires the Digirig product string. Verify with: `udevadm info -a -n /dev/ttyUSB0 | grep -E "idVendor|idProduct|product"`

```
# List all USB serial devices
ls -la /dev/ttyUSB* /dev/ttyACM* 2>/dev/null

# Check the stable symlinks
ls -la /dev/gps0 /dev/tnc0 2>/dev/null

# Confirm GPS is outputting NMEA sentences
sudo cat /dev/gps0

# Use /dev/tnc0 in YAAC → Configure → Ports → Serial TNC
```

## Complete Bill of Materials

| ITEM                                       | MODEL / SPEC                                                                                                                                                                                 | WHERE TO BUY                                    |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| — FieldCommand Server (Pi 5 — 16 GB) —     |                                                                                                                                                                                              |                                                 |
| Raspberry Pi 5 — 16 GB RAM                 | Raspberry Pi 5 Model B (16 GB) — note: DRAM shortage pricing, ~\$305 as of mid-2026 (was \$120)                                                                                              | raspberrypi.com · Adafruit · PiShop.us (~\$305) |
| Pironman MAX 5 enclosure                   | Pironman MAX 5 (tower, dual M.2 NVMe, OLED display, active cooling fans)                                                                                                                     | 52pi.com · Amazon                               |
| NVMe SSD × 2 (RAID 1)                      | 1 TB M.2 2280 PCIe Gen 3/4 NVMe (e.g. WD Blue SN580 or Samsung 980) — NAND shortage pricing ~\$180 each as of mid-2026 (was ~\$75)                                                           | Amazon · B&H; · Newegg (~\$180 each)            |
| Pi 5 USB-C power supply                    | Official Raspberry Pi 27W USB-C PD power supply                                                                                                                                              | raspberrypi.com · Adafruit · Amazon             |
| MicroSD card (boot / initial RAID setup)   | 32 GB Class 10 / A1 microSD (removed after RAID is built)                                                                                                                                    | Amazon                                          |
| — 44Net Gateway (Pi 5 — 16 GB) —           |                                                                                                                                                                                              |                                                 |
| Raspberry Pi 5 — 16 GB RAM                 | Raspberry Pi 5 Model B (16 GB) — dedicated AMPRNet gateway — matches FieldCommand Pi for spare-parts commonality — 8 GB variant (~\$130) also works for gateway-only role if budget is tight | raspberrypi.com · Adafruit · PiShop.us (~\$305) |
| Argon NEO 5 M.2 BRED case                  | Argon NEO 5 M.2 BRED (compact, passive-cooled, M.2 SATA slot, aluminum) — fits Pi 5                                                                                                          | argon40.com · Amazon (~\$35)                    |
| M.2 SATA SSD (gateway OS + config)         | 256 GB M.2 2242 or 2280 SATA SSD — OS, WireGuard config, routing tables                                                                                                                      | Amazon · Newegg (~\$25)                         |
| Pi 5 USB-C power supply                    | Official Raspberry Pi 27W USB-C PD power supply — one per Pi                                                                                                                                 | raspberrypi.com · Adafruit · Amazon             |
| — Networking —                             |                                                                                                                                                                                              |                                                 |
| ASUS RT-BE58 Go travel router              | ASUS RT-BE58 Go (Wi-Fi 7, 2.5G LAN uplink, USB-C 18W power) — qty: 3 recommended (1 primary + 2 AiMesh nodes)                                                                                | Amazon · Best Buy · B&H;                        |
| UniFi Switch Lite 16 PoE                   | Ubiquiti USW-Lite-16-PoE (16-port GbE, 8× PoE, 2× SFP uplink, 45W PoE budget)                                                                                                                | ui.com · Amazon · B&H; (~\$200)                 |
| CAT 6 Ethernet cables × 10                 | 1 ft – 6 ft CAT 6 patch cables — router to switch, both Pis, laptop, printer, 4× workstations                                                                                                | Amazon · Monoprice                              |
| — Accessories — Backup & Connectivity —    |                                                                                                                                                                                              |                                                 |
| USB-A drive (32 GB+, labeled FIELDCOMMAND) | 32 GB+ USB 3.0 drive — auto-backup trigger when inserted                                                                                                                                     | Amazon                                          |

| ITEM                                                               | MODEL / SPEC                                                                                                                                                | WHERE TO BUY                                 |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| External USB hard drive 1 TB+ (e.g. LaCie Rugged)                  | 1 TB+ USB 3.0/USB-C portable drive — label FIELDCOMMAND for auto-backup                                                                                     | B&H; · Amazon · Best Buy                     |
| USB GPS receiver (optional)                                        | GlobalSat BU-353-S4 or u-blox USB GPS puck — provides live position to tactical map and NWS alerts                                                          | Amazon                                       |
| Powered USB hub (optional)                                         | Anker 7-Port USB 3.0 with AC power adapter — must be powered hub, not bus-powered                                                                           | Amazon · Best Buy                            |
| Avery 5371 business card sheets (operator access cards)            | Avery 5371 — 10 cards per sheet, laser or inkjet — for printed operator access cards                                                                        | Office Depot · Amazon · Staples              |
| — Amateur Radio Equipment (skip if no amateur radio group) —       |                                                                                                                                                             |                                              |
| USB TNC — Digirig Mobile                                           | Digirig Mobile v1.x — USB soundcard + serial CAT in one compact unit — enables APRS transmit/receive via Direwolf/YAAC and Winlink VHF packet via Pat       | digirig.net · Amazon                         |
| USB TNC — Signalink USB (alternative)                              | Tigertronics Signalink USB — USB audio interface — requires rig-specific audio cable — widely used and well-supported                                       | tigertronics.com · Ham Radio Outlet · Amazon |
| Mobilinkd TNC4 (Bluetooth alternative)                             | Bluetooth APRS TNC — pairs with YAAC on the Pi or APRSdroid on a phone — useful when a USB port is not available                                            | mobilinkd.com · Amazon                       |
| Rig interface cable (for TNC)                                      | Radio-specific audio + PTT cable connecting TNC to transceiver — Digirig: cables at digirig.net by radio model — Signalink: Tigertronics cable for each rig | digirig.net · tigertronics.com · HRO         |
| Windows laptop (Winlink Express + JS8Call)                         | Any Windows 10/11 laptop with USB-A port and Wi-Fi — required for Winlink Express, VARA HF, and JS8Call — see Part 2E                                       | Best Buy · Amazon                            |
| — Printers — Monochrome Laser (USB via CUPS or direct Wi-Fi) —     |                                                                                                                                                             |                                              |
| Brother HL-L2350DW (recommended)                                   | Monochrome laser — USB + Wi-Fi — excellent Linux driver — fast, duplex, durable                                                                             | Best Buy · Amazon (~\$130)                   |
| HP LaserJet Pro M15w / P1102w                                      | Monochrome laser — USB + Wi-Fi — HPLIP driver auto-installs with CUPS                                                                                       | Best Buy · Amazon (~\$150)                   |
| — Printers — Color Multifunction Laser (USB, Wi-Fi, or Ethernet) — |                                                                                                                                                             |                                              |
| Brother MFC-L3770CDW                                               | Color laser MFP — print, scan, copy, fax — USB + Wi-Fi + Ethernet — full Linux CUPS support — prints color ICS maps and IAP packages                        | Amazon · Best Buy (~\$400)                   |
| HP Color LaserJet Pro MFP M479fdw                                  | Color laser MFP — print, scan, copy, fax — USB + Wi-Fi + Ethernet — HPLIP driver works out of the box with CUPS                                             | Amazon · B&H; (~\$500)                       |

| ITEM                                                                              | MODEL / SPEC                                                                                                               | WHERE TO BUY                                                                                                     |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Canon imageCLASS MF743Cdw                                                         | Color laser MFP — letter + legal — USB + Wi-Fi + Ethernet — Linux CAPT driver — good for permanent EOC installations       | Amazon · Office Depot (~\$450)                                                                                   |
| — Printers — Portable / Battery-Powered (field deployments without shore power) — |                                                                                                                            |                                                                                                                  |
| Canon PIXMA TR150                                                                 | Compact inkjet — built-in rechargeable battery — ~200 pages per charge — USB + Wi-Fi — letter size                         | Best Buy · Amazon (~\$200)                                                                                       |
| HP OfficeJet 200                                                                  | Portable inkjet — larger paper tray — optional battery accessory — USB + Wi-Fi — good for shelter stations                 | Best Buy · Amazon (~\$180)                                                                                       |
| — Operator Workstations — Raspberry Pi 500 / 500+ (qty: 4 recommended) —          |                                                                                                                            |                                                                                                                  |
| Raspberry Pi 500 keyboard computer ×4                                             | Pi 500 standalone — Pi 5 chip, 8 GB RAM, 32 GB A2 microSD, integrated keyboard — one per operator station                  | raspberrypi.com · Adafruit · Amazon (~\$130 each — DRAM shortage pricing)                                        |
| Raspberry Pi 500+ ×4 (alternative)                                                | Pi 500+ — same as Pi 500 but with Pi 5 latest revision chip and 2× USB3 — specify if 501(c)(3) funding allows premium spec | raspberrypi.com · Adafruit (~\$410 each — significantly increased by DRAM shortage; standard Pi 500 recommended) |
| Raspberry Pi Monitor 15.6" ×4                                                     | 15.6" Full HD IPS touchscreen — built-in speakers — kickstand — VESA mount — USB-C powered from Pi 500 — one per station   | raspberrypi.com · GigaParts · Amazon (~\$100 each)                                                               |
| micro-HDMI to HDMI cable ×4                                                       | 1m micro-HDMI to standard HDMI — Pi 500 to Monitor video connection — one per station                                      | Included in Pi 500 Desktop Kit · Amazon (~\$8 each)                                                              |
| USB-C power supply for Pi 500 ×4                                                  | Official Raspberry Pi 27W USB-C PD supply — one per station — Pi 500 can also be powered from Monitor USB-C port           | raspberrypi.com · Adafruit (~\$12 each)                                                                          |
| Raspberry Pi 500 Desktop Kit ×4 (bundle option)                                   | Pi 500 + official mouse + micro-HDMI cable bundled — add Monitor separately — simpler ordering than individual parts       | raspberrypi.com · Amazon (~\$160 each — DRAM shortage pricing)                                                   |

**Amateur Radio Equipment category:** These items are only needed if a licensed amateur radio group is the lead or a key partner in the deployment. If FieldCommand is deployed by a public safety agency or served organization without licensed amateur radio operators, skip the entire Amateur Radio Equipment category. See Part 2 of this guide for amateur radio setup procedures.

## EMCOMM-NET Coverage Extension — 3× ASUS RT-BE58 Go Total (1 Primary + 2 AiMesh Nodes)

The standard FieldCommand deployment uses three ASUS RT-BE58 Go routers: one configured as the primary (connected to your WAN source and the UniFi switch) and two configured as AiMesh nodes to extend EMMCOMM-NET coverage across large EOC buildings, shelters, and outdoor staging areas. All three are the same hardware — any unit can serve as the primary if needed. All three broadcast the same EMMCOMM-NET SSID and password — devices roam between them automatically with no reconnection required.

| ITEM                                     | MODEL / SPEC                                                                                                                                                                                            | WHERE TO BUY                                    |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| ASUS RT-BE58 Go ×2 (mesh nodes)          | ASUS RT-BE58 Go — identical to primary router — pairs instantly as AiMesh node with no extra configuration — extends EMCOMM-NET at full Wi-Fi 7 speed — wired backhaul via UniFi switch ports 11 and 12 | Amazon · Best Buy · B&H; (~\$120-150 each)      |
| USB-C power supply ×2 (for mesh nodes)   | USB-C 18W PD adapter — one per node — or USB-C power bank (10,000+ mAh, 18W+) for field use without shore power                                                                                         | Amazon (~\$15-25 each)                          |
| CAT 6 Ethernet cable ×2 (wired backhaul) | CAT 6 patch cable from UniFi Switch Lite 16 PoE to each node LAN port — wired backhaul is strongly recommended — provides full throughput vs wireless backhaul — length as needed for the venue         | Amazon · Monoprice · Home Depot (~\$10-20 each) |

| DEPLOYMENT SCENARIO                                 | RECOMMENDED SETUP                                                               |
|-----------------------------------------------------|---------------------------------------------------------------------------------|
| Single room EOC (under 2,500 sq ft)                 | 1× RT-BE58 Go primary only — no extension needed                                |
| Multi-room EOC or large shelter (2,500–7,500 sq ft) | 1× primary + 1 RT-BE58 Go node — wired backhaul via switch Port 11              |
| Large building or campus (7,500–20,000 sq ft)       | 1× primary + 2 RT-BE58 Go nodes — wired backhaul via switch Ports 11 and 12     |
| Outdoor SAR staging area                            | 1× primary at command post + 1-2 RT-BE58 Go nodes at field positions on battery |

■ All three ASUS RT-BE58 Go routers broadcast the same EMCOMM-NET SSID and password. Operators and devices roam between them automatically — no reconnection, no password re-entry, no awareness of which router they are connected to. Pairing a new node to the primary takes under 5 minutes on site.

## SD Card / Storage Sizing Guide

| TIER                | WHAT FITS / REQUIREMENT                                             |
|---------------------|---------------------------------------------------------------------|
| Minimum (boot only) | 8 GB — OS only, RAID boot loader. FieldCommand runs from NVMe RAID. |
| Recommended         | 32 GB — OS + swap + logs + temp files during initial RAID setup.    |
| Kiwix Tier 1        | ~3 GB additional on RAID — WikiMed + Wikipedia Mini + Wikivoyage.   |
| Kiwix Tier 2        | ~10 GB additional on RAID — adds iFixit, Wikibooks, Wikiversity.    |
| FCC Database        | ~600 MB on RAID — full US amateur license database (~800K records). |

# 1 Flash OS & Build RAID 1 — Do This First

## Step 1: Flash the OS and Build the RAID Array

These two tasks must be completed before anything else. The Raspberry Pi OS must be installed and the RAID 1 array assembled before the FieldCommand installer runs. If you are using a single NVMe drive (no RAID), flash the OS to it directly and skip the RAID build section.

### 1A Flash the Operating System

| OS EDITION                                                            | PROS                                                                                            | CONS                                                                                                                                                   |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Raspberry Pi OS Lite (64-bit) — Recommended for production            | Minimal overhead. All RAM available to FieldCommand. Fast boot. Smaller attack surface.         | No local browser or GUI — configuration via SSH from your laptop. FieldCommand itself is always accessed from EMCOMM-NET devices, not the Pi directly. |
| Raspberry Pi OS Desktop (64-bit) — Good for setup and troubleshooting | Local browser for testing. GUI for initial Wi-Fi setup and configuration. Easier for new users. | More RAM used by desktop environment. Slightly longer install time.                                                                                    |
| Ubuntu Server 24.04 LTS — Also supported                              | LTS kernel. Familiar to Linux admins. Long-term package support.                                | Some package names differ from Raspberry Pi OS. Slightly more manual setup.                                                                            |

Use **Raspberry Pi Imager** (download from [raspberrypi.com/software](https://raspberrypi.com/software)) to flash the OS to either a microSD card (for single-drive setup) or directly to the NVMe SSD (if not using RAID). During the Imager advanced options, set:

| IMAGER OPTION     | VALUE / REASON                                                                                                                                                                               |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hostname          | fieldcommand.local — makes the Pi discoverable as fieldcommand.local on the network                                                                                                          |
| Username          | fieldcommand — the FieldCommand installer expects this username                                                                                                                              |
| Password          | Choose a strong password and record it — required for SSH and CUPS admin                                                                                                                     |
| Enable SSH        | <b>Lite:</b> Yes — required (only access method without a display). <b>Desktop:</b> Recommended for remote diagnostics, but optional if you have a monitor and keyboard connected to the Pi. |
| Wi-Fi             | Set your home or lab Wi-Fi during initial OS setup. EMCOMM-NET comes later.                                                                                                                  |
| Locale / Timezone | US/Central for local operations                                                                                                                                                              |

### 1B Build the RAID 1 Array (Pironman MAX 5 — Dual NVMe)

The Pironman MAX 5 enclosure has two M.2 NVMe slots. Configuring them as a RAID 1 mirror means both drives always contain identical data. If one SSD fails, the Pi keeps running on the surviving drive with no data loss and no operator action required. Skip this section entirely if you are using a single NVMe drive.



Complete the RAID build BEFORE running the FieldCommand installer. The installer must be run on the final storage configuration. If your OS is already running from a single drive, back it up before proceeding.

### Physical Installation

| SLOT                 | DRIVE                              | LINUX DEVICE NAME |
|----------------------|------------------------------------|-------------------|
| M.2 Slot 1 (primary) | 1 TB NVMe SSD (e.g. WD Blue SN580) | /dev/nvme0n1      |

| SLOT                | DRIVE                                       | LINUX DEVICE NAME |
|---------------------|---------------------------------------------|-------------------|
| M.2 Slot 2 (mirror) | 1 TB NVMe SSD — identical model recommended | /dev/nvme1n1      |

Boot from a microSD card with Raspberry Pi OS Lite to perform the initial RAID setup. After the array is built and the OS copied to it, the SD card is removed.

### Build the Array

```
# Install mdadm (RAID management tool)
sudo apt-get update && sudo apt-get install -y mdadm

# Create the RAID 1 array (--run skips the "all zeroes" check for speed)
sudo mdadm --create --verbose /dev/md0 \
--level=1 --raid-devices=2 \
/dev/nvme0n1 /dev/nvme1n1

# Check build progress (shows resync percentage – takes ~15 min for 1 TB)
cat /proc/mdstat

# Create a filesystem on the array
sudo mkfs.ext4 -L fieldcommand-raid /dev/md0

# Mount and copy the running OS to the RAID array
sudo mkdir -p /mnt/raid
sudo mount /dev/md0 /mnt/raid
sudo apt-get install -y rsync
sudo rsync -axv --progress / /mnt/raid/

# Save the RAID config so it survives reboots
sudo mdadm --detail --scan | sudo tee -a /mnt/raid/etc/mdadm/mdadm.conf
sudo update-initramfs -u -k all

# Update /boot/cmdline.txt to boot from the RAID array instead of the SD
sudo nano /boot/cmdline.txt
# Change: root=PARTUUID=xxxx → root=/dev/md0
# Then reboot – Pi boots from RAID. Verify with:
cat /proc/mdstat
# Expected: md0 : active raid1 nvme0n1[0] nvme1n1[1] [UU]
```

| RAID STATUS                     | MEANING                                                             | ACTION REQUIRED                                                                                                            |
|---------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| [UU] (both drives active)       | Normal — both drives healthy and in sync                            | None                                                                                                                       |
| [U_] or [_U] (one drive failed) | One drive has failed. Pi is still running on the other.             | Power down. Replace failed drive. Run: <code>sudo mdadm /dev/md0 --add /dev/nvme1n1</code> . Array rebuilds automatically. |
| Resync in progress              | Array is rebuilding — normal after first setup or drive replacement | Leave running. Takes 15 – 30 minutes. Pi is usable during rebuild.                                                         |



RAID 1 protects against drive failure — it is NOT a backup. Both drives contain identical data, so accidental deletion affects both drives simultaneously. Continue to use the FIELDCOMMAND USB auto-backup for regular data backups.

## 2. Before You Begin — Prerequisites

Complete these prerequisites before running the installer. The installer requires an internet connection for package downloads, the FCC database, and Kiwix ZIM files. After installation, the server operates fully offline.

### 2.1 Flash the Operating System

| OS EDITION                                                            | PROS                                                                                              | CONS                                                                                                                                        |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Raspberry Pi OS Lite (64-bit) — Recommended for production            | Minimal overhead, faster boot, all RAM available to FieldCommand services, smaller attack surface | No local browser or GUI — configuration via SSH from your laptop. FieldCommand is accessed from EMCOMM-NET devices, not from the Pi itself. |
| Raspberry Pi OS Desktop (64-bit) — Good for setup and troubleshooting | GUI for initial setup, Raspberry Pi Configuration tool, local browser for testing                 | More RAM used by desktop environment; install takes longer                                                                                  |
| Ubuntu Server 24.04 LTS — Also supported                              | LTS kernel, familiar to Linux admins, good for organizations standardized on Ubuntu               | Slightly larger memory footprint; some package names differ                                                                                 |

### 2.2 First Boot — Initial Setup

| STEP | ACTION                                                                                                                                                                                                     |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | On first boot, complete the initial setup wizard: set username to <b>fieldcommand</b> , set a strong password, configure locale and keyboard, expand filesystem.                                           |
| 2    | If using Desktop edition: complete the initial desktop setup wizard.                                                                                                                                       |
| 3    | <b>Lite only — Enable SSH:</b> run <code>sudo raspi-config</code> → Interface Options → SSH → Enable. <b>Desktop users skip this step</b> — use the local terminal instead of SSH for all remaining steps. |

Run a full system update before installing FieldCommand (works the same on both editions — Terminal or SSH):

```
# Update all packages to latest versions
sudo apt-get update && sudo apt-get full-upgrade -y
# Reboot to apply kernel updates
sudo reboot
```

### 2.3 Verify Disk Space

```
# Check available space – should show 20GB+ free for Tier 2 install
df -h /
# Verify RAM
free -h
```

### 2.4 Internet Connection

Connect the Pi to the internet via Ethernet before running the installer. Wi-Fi can be used during initial setup but Ethernet is strongly preferred for downloading large files — the FCC database is ~600 MB and Kiwix ZIMs range from 500 MB to 25 GB.

### 2.5 Operator Workstation — Browser Options

FieldCommand runs entirely in a web browser. Any modern browser on any device connected to EMCOMM-NET works. For the Raspberry Pi 500 / 500+ used as an operator workstation, three browsers are available:

| BROWSER              | HOW TO INSTALL                                      | NOTES FOR FIELDCOMMAND                                                                                                                                                                           |
|----------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chromium (default)   | Pre-installed on Raspberry Pi OS. No action needed. | Best performance for FieldCommand. Discovers CUPS printers automatically via Bonjour. Set <a href="http://192.168.50.1">http://192.168.50.1</a> as the startup page under Settings → On startup. |
| Firefox ESR          | <code>sudo apt install firefox-esr</code>           | Excellent compatibility with all FieldCommand pages. Good fallback if Chromium behaves unexpectedly. Also discovers network printers. Slightly lower memory footprint on the Pi 500.             |
| GNOME Web (Epiphany) | <code>sudo apt install epiphany-browser</code>      | Very lightweight WebKit browser. Good for Pi 500 with many tabs open. Best for dedicated check-in stations where memory is tight.                                                                |



For dedicated FieldCommand operator stations running Pi 500: set Chromium to open <http://192.168.50.1> on startup. In Chromium: Settings → On startup → Open a specific page → enter <http://192.168.50.1>. The dashboard loads automatically every time the Pi 500 boots.



Wi-Fi is provided by the ASUS RT-BE58 Go router — the Pi does NOT run hostapd. During installation, connect the Pi to the internet via Ethernet through the UniFi switch and ASUS router. After installation, the Pi is always reachable at [192.168.50.1](http://192.168.50.1) via wired Ethernet.

## 2

## Network Hardware Setup — ASUS Router + UniFi Switch

## Step 2: Network Hardware Setup — ASUS Router + UniFi Switch

Complete this step before running the FieldCommand installer. The ASUS router must be configured with the correct LAN subnet (192.168.50.x) before FieldCommand is reachable at <http://192.168.50.1>.

## Physical Wiring

The UniFi Switch Lite 16 PoE is the central wiring hub for the entire EMCOMM-NET deployment. Connect everything to the switch, and the switch connects up to the ASUS router. Use short CAT 6 patch cables (1 ft to 3 ft) to keep the rack or case tidy. The primary cellular modem or antenna connects directly to the ASUS WAN port via its PoE Ethernet cable — this is completely separate from the switch and does not use a switch port.

| CONNECTION                          | DEVICE                                 | IP / NOTES                                                                                                                                                             |
|-------------------------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASUS WAN port (PoE Ethernet)        | Cellular antenna (primary WAN)         | Primary internet — PoE cable from outdoor cellular modem/antenna directly to ASUS WAN port. Cellular modem serves DHCP on its own subnet — ASUS bridges to EMCOMM-NET. |
| ASUS USB WAN port (USB-to-Ethernet) | Satellite dish (secondary WAN)         | Secondary internet — satellite. satellite Ethernet adapter → USB-to-Ethernet adapter → ASUS USB port. ASUS switches automatically when cellular WAN drops.             |
| Switch Port 1 (uplink)              | ASUS RT-BE58 Go LAN port               | Uplink from router to switch. Router is DHCP server for 192.168.50.0/24.                                                                                               |
| Port 2                              | FieldCommand Pi 5 (Pironman MAX 5)     | 192.168.50.1 — All browser tools live here.                                                                                                                            |
| Port 3                              | 44Net Gateway Pi 5 (Argon NEO 5)       | 192.168.50.2 — AMPRNet WireGuard gateway.                                                                                                                              |
| Port 4                              | Windows Laptop (Winlink + JS8Call)     | 192.168.50.3 (static reservation) — or via EMCOMM-NET Wi-Fi.                                                                                                           |
| Port 5                              | Color MFP Network Printer              | 192.168.50.10 (recommended static reservation) — auto-discovered by all devices via Bonjour.                                                                           |
| Ports 6 – 9                         | Pi 500 Operator Workstations (up to 4) | 192.168.50.20 – 192.168.50.23 (static DHCP reservations)                                                                                                               |
| Port 10                             | Satellite dish router (optional)       | Optional — connect satellite Ethernet to switch as WAN source on ASUS router.                                                                                          |
| Port 11                             | ASUS RT-BE58 Go — Mesh Node 1          | Wired backhaul to first mesh node. Extends EMCOMM-NET to secondary coverage zone.                                                                                      |
| Port 12                             | ASUS RT-BE58 Go — Mesh Node 2          | Wired backhaul to second mesh node. Extends EMCOMM-NET to third coverage zone.                                                                                         |
| Ports 13 – 16                       | Spare                                  | Available for additional devices, IP radios, second TNC, or future expansion.                                                                                          |

| DEVICE                 | POWER SOURCE                                             |
|------------------------|----------------------------------------------------------|
| ASUS RT-BE58 Go router | USB-C 18W PD adapter — or USB-C power bank for field use |

| DEVICE                             | POWER SOURCE                                                             |
|------------------------------------|--------------------------------------------------------------------------|
| UniFi Switch Lite 16 PoE           | Included 45W AC power supply                                             |
| FieldCommand Pi 5 (Pironman MAX 5) | Official Raspberry Pi 27W USB-C PD supply via Pironman MAX 5 power inlet |
| 44Net Gateway Pi 5 (Argon NEO 5)   | Official Raspberry Pi 27W USB-C PD supply directly to Pi USB-C port      |

## Configure the ASUS RT-BE58 Go

| STEP | ACTION                                                                                                                                                                                                                  |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Power on the router. Connect a device to it via Wi-Fi (default SSID on label) or Ethernet. Open <a href="http://192.168.50.1">http://192.168.50.1</a> (or the default router IP on the label).                          |
| 2    | Complete the router setup wizard. When it asks for WAN, choose Automatic IP (DHCP) or skip if no internet is connected.                                                                                                 |
| 3    | Go to <b>LAN</b> → <b>LAN IP</b> . Set the LAN IP to <b>192.168.50.1</b> , subnet <b>255.255.255.0</b> . Apply.                                                                                                         |
| 4    | Go to <b>LAN</b> → <b>DHCP Server</b> . Set IP pool: <b>192.168.50.100 – 192.168.50.200</b> . Apply.                                                                                                                    |
| 5    | Change the ASUS router admin password to something strong. Record it on a piece of paper stored on the FIELDCOMMAND USB drive.                                                                                          |
| 6    | Set Manually Assigned IP for the Pi: go to <b>LAN</b> → <b>DHCP Server</b> → <b>Manually Assigned IP</b> . Enter the Pi's MAC address → assign IP <b>192.168.50.1</b> . (Or set the Pi's static IP manually in Step 4.) |
| 7    | Set the Wi-Fi SSID: go to <b>Wireless</b> → <b>General</b> . Set both the 2.4 GHz and 5 GHz SSIDs to <b>EMCOMM-NET</b> . Set a strong WPA3/WPA2 password. Apply.                                                        |

The ASUS RT-BE58 Go supports dual WAN with automatic failover between a primary and secondary internet source. When both WAN sources are unavailable, FieldCommand continues operating on all local features without interruption — net logging, ICS platform, APRS, and all tools remain fully functional.

FieldCommand supports any WAN source your ASUS router can connect to: cellular modem or antenna (PoE Ethernet to WAN port), satellite dish (via Ethernet adapter to USB WAN), site Ethernet, USB smartphone tether, or venue Wi-Fi (WISP mode). Configure your specific WAN hardware according to that device's documentation, then connect it to the ASUS router as shown in the table below. See the ESV Installation Guide for specific WAN hardware configuration.

| INTERNET SOURCE                          | ASUS CONNECTION                                 | HOW TO CONNECT                                                                                                                                                                                  | ROLE                                                                              |
|------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Cellular modem or antenna (PoE Ethernet) | WAN Ethernet port                               | Run PoE Ethernet cable from the outdoor modem/antenna directly to the ASUS WAN port. Set WAN type → Automatic IP (DHCP). The modem serves DHCP on its own subnet — ASUS bridges to EMMCOMM-NET. | Primary or secondary — your group chooses based on available equipment            |
| Satellite dish (Ethernet adapter)        | ASUS USB WAN port (via USB-to-Ethernet adapter) | Satellite Ethernet adapter → USB-to-Ethernet adapter → ASUS USB port. Enable Dual WAN in ASUS admin and configure failover order.                                                               | Primary or secondary — your group chooses based on available equipment            |
| EOC or site wired Ethernet               | WAN Ethernet port                               | Plug site Ethernet cable into the ASUS WAN port. Set WAN type → Automatic IP (DHCP).                                                                                                            | Good primary when site has reliable wired internet — low cost, no hardware needed |

| INTERNET SOURCE         | ASUS CONNECTION   | HOW TO CONNECT                                                                                                     | ROLE                                                                 |
|-------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| USB smartphone tether   | ASUS USB WAN port | Enable hotspot on phone. Connect USB cable from phone to the ASUS USB port. Set WAN → USB Tethering in ASUS admin. | Last resort — when no other WAN source is available at the site      |
| Venue Wi-Fi (WISP mode) | ASUS wireless WAN | ASUS admin → Operation Mode → Wireless ISP (WISP). Select the venue Wi-Fi network and enter the password.          | When the site has existing guest Wi-Fi — hospital, shelter, or venue |

## AiMesh Mesh Node Setup — Extending EMCOMM-NET Coverage

The standard FieldCommand deployment includes three ASUS RT-BE58 Go routers: one primary (connected to the WAN source and the UniFi switch) and two mesh nodes that extend EMCOMM-NET to additional rooms, floors, or outdoor areas. All three use the same hardware, the same SSID, and the same password. Devices roam between them automatically.

■ Connect the two mesh nodes via wired Ethernet backhaul (switch Ports 11 and 12) rather than wireless backhaul whenever possible. Wired backhaul provides full Wi-Fi 7 throughput on both upstream and downstream — wireless backhaul cuts bandwidth roughly in half because the node uses its radios for both the backhaul link and the client connections simultaneously.

**Repeat these steps for each of the two mesh nodes:**

| STEP | ACTION                                                                                                                                                                                                                                              |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>Connect the node Ethernet</b> — run a CAT 6 cable from UniFi Switch Port 11 (first node) or Port 12 (second node) to the LAN port of the mesh node router. This is the wired backhaul — it carries all traffic between the node and the primary. |
| 2    | <b>Power on the node</b> — connect the USB-C power supply to the mesh node. For initial pairing, position the node within 30 feet of the primary router. After pairing it can be moved to its final location.                                       |
| 3    | <b>Open AiMesh on the primary router</b> — go to <a href="http://192.168.50.254">http://192.168.50.254</a> → <b>AiMesh</b> → <b>AiMesh Node</b> . The new node appears in the device list within 1-2 minutes.                                       |
| 4    | <b>Click Connect</b> on the new node. The primary router pushes the EMCOMM-NET SSID, password, and configuration to the node automatically. This takes 1-3 minutes.                                                                                 |
| 5    | <b>Move the node to its final position</b> — mount or place it in the coverage area it needs to serve. The wired backhaul cable connects back to the UniFi switch. Test coverage with a phone on EMCOMM-NET at the furthest point in the area.      |
| 6    | <b>Verify in AiMesh dashboard</b> — on the primary router, AiMesh → AiMesh Node should show both nodes as Connected with backhaul type Ethernet. Green status on both nodes confirms the mesh is fully operational.                                 |

| NODE                                   | SWITCH PORT                  | RECOMMENDED PLACEMENT                                                |
|----------------------------------------|------------------------------|----------------------------------------------------------------------|
| Primary RT-BE58 Go (AiMesh controller) | Uplink to ASUS 2.5G LAN port | Central to the deployment — EOC command post or entrance area        |
| RT-BE58 Go — Mesh Node 1               | UniFi Switch Port 11         | Secondary room, opposite wing, or upper floor                        |
| RT-BE58 Go — Mesh Node 2               | UniFi Switch Port 12         | Third coverage zone — outdoor staging, parking, or far building wing |

## 3 Download & Run the FieldCommand Installer

### Step 3: Download & Run the FieldCommand Installer

#### Method A — Direct download to the Pi (requires internet on Pi)

**Lite:** SSH into the Pi and run these commands. **Desktop:** open a Terminal window (taskbar → Terminal) and run the same commands.

```
# Create working directory
mkdir -p ~/fieldcommand-install && cd ~/fieldcommand-install
# Download the FieldCommand package from GitHub or your storage location
wget -O fieldcommand-v1.zip
https://github.com/KE4CON/FieldCommand-IMS/releases/latest/download/fieldcommand-v1.zip
# Unzip the package
unzip fieldcommand-v1.zip
cd fieldcommand
```

#### Method B — Copy from your computer using SCP (Lite/headless)

```
# Run this on YOUR COMPUTER, not the Pi
scp fieldcommand-v1.zip fieldcommand@[pi-ip-address]:~/
# Then SSH into the Pi and unzip
ssh fieldcommand@[pi-ip-address]
mkdir -p ~/fieldcommand-install
mv fieldcommand-v1.zip ~/fieldcommand-install/
cd ~/fieldcommand-install && unzip fieldcommand-v1.zip
cd fieldcommand
```

#### Method B2 — Download directly on the Pi Desktop (Desktop edition)

| OPTION           | HOW                                                                                                                                             |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Browser download | Open Chromium on the Pi. Go to the GitHub releases page or download URL. Download fieldcommand-v1.zip — it saves to ~/Downloads/ automatically. |
| USB drive        | Copy fieldcommand-v1.zip to a USB drive. Insert the USB into the Pi. Open File Manager, drag the zip to your home folder.                       |

```
# If downloaded to ~/Downloads/:
mkdir -p ~/fieldcommand-install
cp ~/Downloads/fieldcommand-v1.zip ~/fieldcommand-install/
# Unzip (same for all methods):
cd ~/fieldcommand-install && unzip fieldcommand-v1.zip
cd fieldcommand
```

#### Method C — USB drive transfer (fully offline)

**Lite:** commands below via SSH. **Desktop:** use File Manager to copy the zip from USB to your home folder, then open a Terminal for the unzip step.

```
# Copy fieldcommand-v1.zip to a USB drive on your computer
# Insert USB drive into the Pi, then:
sudo mount /dev/sda1 /mnt # or check lsblk for your device
cp /mnt/fieldcommand-v1.zip ~/fieldcommand-install/
cd ~/fieldcommand-install && unzip fieldcommand-v1.zip
cd fieldcommand
```

## Launch the Installer

**Lite:** run via SSH. **Desktop:** open a Terminal window from the taskbar or Accessories → Terminal. The installer runs identically on both.

```
cd ~/fieldcommand-install/fieldcommand
sudo bash scripts/install.sh
```

## 4 Installer Configuration Options

### Step 4: Installer Configuration Options

| PROMPT                | DEFAULT          | DESCRIPTION                                                                               |
|-----------------------|------------------|-------------------------------------------------------------------------------------------|
| Station callsign      | W8ABC            | Your amateur callsign. Patched into all HTML pages.                                       |
| Station latitude      | 42.3153          | Decimal degrees. Used for APRS station marker, NWS alerts, propagation, distance calc.    |
| Station longitude     | -88.4473         | Decimal degrees (negative = West). Default 0.0/0.0 — set to your deployment coordinates.  |
| Wi-Fi AP SSID         | EMCOMM-NET       | The Wi-Fi network name that field devices connect to.                                     |
| Wi-Fi AP password     | fieldcommand2026 | WPA2 password for EMMCOMM-NET. Change this for operational security.                      |
| Server IP address     | 192.168.50.1     | The Pi's static IP on EMMCOMM-NET. Devices browse to http://192.168.50.1/                 |
| Download FCC database | N                | ~600 MB download of the FCC amateur license database. Recommended: Y                      |
| Kiwix tier            | 1                | Tier 1: WikiMed + Wikipedia Mini + Wikivoyage (~2.5 GB). Tier 2 adds iFixit (~5 GB more). |

#### What the Installer Does Automatically

- Installs system packages: Python 3, nginx, kiwix-tools, rsync, gpsd, ufw, mdadm, java
- Creates the **fieldcommand** system user and directory structure at /opt/fieldcommand/
- Creates a Python virtual environment and installs Flask, flask-cors, requests, gpsd-py3
- Deploys all 30 HTML pages to /opt/fieldcommand/html/ and patches your callsign and coordinates into each file
- Installs all 14 systemd service and timer units and enables them at boot
- Configures nginx with the correct web root (/opt/fieldcommand/html/) and proxy rules for all API ports
- **Does NOT configure a Wi-Fi access point** — Wi-Fi is handled by the ASUS RT-BE58 Go router. The Pi connects as a wired client only.
- Configures the Pi static IP (192.168.50.1/24) on the Ethernet interface (eth0) via NetworkManager
- Configures the firewall (ufw) — opens all required ports
- Installs Direwolf (via apt) as the software TNC and YAAC (Java APRS client, /opt/yaac/)
- Installs the USB backup udev rule (plug in a USB drive labeled FIELDCOMMAND to trigger auto-backup)
- Downloads and installs Pat Winlink (browser-based Winlink backup client, port 8090)
- Runs the Kiwix ZIM downloader for the selected content tier
- Downloads the FEMA ICS forms PDFs to /opt/fieldcommand/data/ics\_forms/ (22 forms)
- Optionally downloads and builds the FCC amateur license database (~600 MB)
- Starts all services and verifies they are running



The ASUS RT-BE58 Go router must be configured before running the installer — see Step 1. The Pi does NOT run hostapd or dnsmasq. All Wi-Fi, DHCP, and SSID management is handled by the ASUS router.

## 4b

## RAID 1 NVMe Setup

## Step 4b: RAID 1 NVMe Setup — Pironman MAX 5

The Pironman MAX 5 enclosure has two M.2 NVMe slots. Configure them as a RAID 1 mirror so both drives contain identical data. If one drive fails, the system continues running on the surviving drive with no data loss and no operator action required.



Complete this step before running the FieldCommand installer. The RAID array must be assembled and the OS installed on it before FieldCommand is deployed. If your Pi OS is already on a single drive, back it up first.

## Physical Installation

| STEP | ACTION                                                                                                                                |
|------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Install both 1 TB NVMe SSDs into the Pironman MAX 5 M.2 slots. Slot 1 (primary) → /dev/nvme0n1 · Slot 2 (mirror) → /dev/nvme1n1       |
| 2    | Boot from a Raspberry Pi OS microSD card for the initial RAID setup. After RAID is built and OS is installed, the SD card is removed. |

## Build the RAID 1 Array

```
# Install mdadm (RAID management tool)
sudo apt-get install -y mdadm
# Create RAID 1 array from the two NVMe drives
sudo mdadm --create --verbose /dev/md0 --level=1 --raid-devices=2 \
/dev/nvme0n1 /dev/nvme1n1
# Confirm array is building (will show [=>.....] resync progress)
cat /proc/mdstat
# Create a filesystem on the RAID array
sudo mkfs.ext4 -L fieldcommand-raid /dev/md0
# Mount it
sudo mkdir -p /mnt/raid
sudo mount /dev/md0 /mnt/raid
# Copy running OS to RAID array
sudo apt-get install -y rsync
sudo rsync -axv --progress / /mnt/raid/
# Save RAID configuration so it persists across reboots
sudo mdadm --detail --scan | sudo tee -a /mnt/raid/etc/mdadm/mdadm.conf
```

## RAID Health Reference

| SITUATION                      | WHAT TO DO                                                                                                                                                                            |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Both drives show [UU]          | Normal healthy state. No action needed.                                                                                                                                               |
| One drive fails — [_U] or [U_] | 1. Power down. 2. Remove failed drive. 3. Insert new 1 TB NVMe. 4. Power on. 5. Run: <code>sudo mdadm /dev/md0 --add /dev/nvme1n1</code> . 6. Array rebuilds automatically (~30 min). |
| Check RAID health anytime      | <code>cat /proc/mdstat</code> or <code>sudo mdadm --detail /dev/md0</code>                                                                                                            |



RAID 1 protects against drive failure — it is NOT a backup. Both drives contain identical data, so accidental deletion or corruption affects both drives simultaneously. Continue to use the USB auto-backup (insert FIELDCOMMAND USB drive) for regular data backups.

## 5 Raspberry Pi 5 — Static IP Configuration

### Step 5: Raspberry Pi 5 — Static IP Configuration

The Pi must always be reachable at 192.168.50.1. If the installer already set this via a DHCP reservation in the ASUS router, verify with `ip addr show eth0` and skip to Step 5 if 192.168.50.1/24 is shown.

**Choose the path that matches your OS edition:** Path A is for Raspberry Pi OS Lite users who configure the Pi over SSH from their laptop. Path B is for Raspberry Pi OS Desktop users who have a monitor and keyboard connected — use the Pi's own terminal, no SSH required. If you used `raspi-config` during OS setup, use Path C.

#### Path A — Raspberry Pi OS Lite (via SSH from your laptop)

```
# SSH into the Pi (use the IP shown in your router DHCP table)
ssh fieldcommand@[current-pi-ip]
# Find the Ethernet connection name
nmcli con show
# Usually: "Wired connection 1" or "Preconfigured Connection 1"
# Set static IP
sudo nmcli con mod "Wired connection 1" ipv4.addresses 192.168.50.1/24 ipv4.method manual
ipv4.method manual
# Apply the change
sudo nmcli con up "Wired connection 1"
# Verify — should show inet 192.168.50.1/24
ip addr show eth0
```

#### Path B — Raspberry Pi OS Desktop (via Terminal on the Pi)

| STEP | ACTION                                                                                                           |
|------|------------------------------------------------------------------------------------------------------------------|
| 1    | Click the <b>network icon</b> in the taskbar (top-right corner).                                                 |
| 2    | Click <b>Advanced Options</b> → <b>Edit Connections</b> .                                                        |
| 3    | Select your <b>Wired connection</b> and click the gear/edit icon.                                                |
| 4    | Click the <b>IPv4 Settings</b> tab.                                                                              |
| 5    | Change <b>Method</b> from "Automatic (DHCP)" to <b>Manual</b> .                                                  |
| 6    | Click <b>Add</b> and enter: Address: 192.168.50.1 · Netmask: 255.255.255.0 · Gateway: (leave blank)              |
| 7    | Click <b>Save</b> , then close the Network Connections window.                                                   |
| 8    | Disconnect and reconnect the Ethernet cable, or reboot the Pi.                                                   |
| 9    | Open the <b>Terminal</b> and verify: <code>ip addr show eth0</code> — should show <b>inet 192.168.50.1/24</b> .  |
| 10   | Open <b>Chromium</b> on the Pi and go to <code>http://192.168.50.1</code> to confirm FieldCommand loads locally. |

Alternatively, open a Terminal on the desktop and use the same `nmcli` commands shown in Path A. The terminal method works identically on the Desktop edition.

#### Path C — `raspi-config` (if available)

```
# Open raspi-config:
sudo raspi-config
# Navigate to: System Options → Network → (set static IP if option is present)
# NOTE: Static IP configuration via raspi-config was removed in Raspberry Pi OS Bookworm.
# If the option is not visible, use Path A (nmcli) or Path B (GUI Network Connections).
```

## 6

## Kiwix Offline Library Setup

## Step 6: Kiwix Offline Library Setup

Kiwix provides an offline web library served on port 8081. All devices on EMCOMM-NET can browse it at <http://192.168.50.1:8081>. Content is packaged as ZIM files.

| TIER      | NAME                               | SIZE    | DESCRIPTION                                                                                               |
|-----------|------------------------------------|---------|-----------------------------------------------------------------------------------------------------------|
| Essential | WikiMed — Medical Encyclopedia     | ~471 MB | Offline medical encyclopedia: symptoms, treatments, drugs, procedures. Critical for mass-casualty events. |
| Essential | Wikipedia (Mini)                   | ~1.2 GB | Compact English Wikipedia — essential facts, geography, science.                                          |
| Essential | Wikivoyage                         | ~820 MB | Emergency shelters, evacuation routes, local resources and travel logistics.                              |
| Extended  | Wikibooks — How-To & Field Manuals | ~4.2 GB | First aid, survival, ham radio, electronics, cooking, field skills.                                       |
| Extended  | iFixit — Equipment Repair Manuals  | ~3.1 GB | Step-by-step repair guides for electronics, tools, vehicles.                                              |
| Extended  | Wikipedia (Full English)           | ~22 GB  | Complete English Wikipedia. Large download — requires 30 GB+ free.                                        |

```
# Check what is installed and service status
sudo bash /opt/fieldcommand/scripts/kiwix_setup.sh --status

# Add Tier 2 content (resumes interrupted downloads automatically)
sudo bash /opt/fieldcommand/scripts/kiwix_setup.sh --tier 2

# Service management
sudo systemctl status kiwix
sudo systemctl restart kiwix # required after adding new ZIM files
```

ZIM downloads are resumable. If a download is interrupted by power loss or network drop, simply re-run `kiwix_setup.sh` with the same `--tier` flag. `curl` resumes from where it stopped — no need to start over.

### Additional Kiwix Content — Install Any ZIM File

Kiwix.org ([kiwix.org/en/content/](http://kiwix.org/en/content/)) publishes hundreds of ZIM archives beyond the two built-in tiers. Any ZIM file can be added to FieldCommand. The following table lists the most useful additions for emergency communications and public safety operations:

| CONTENT                                 | APPROX SIZE                        | BEST FOR                                                                                         |
|-----------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------|
| OpenStreetMap (Wikimedia Maps)          | ~500 MB                            | Offline detailed maps for any region. Download your county or state for SAR and shelter mapping. |
| Stack Overflow (programming Q&A;)       | ~3.5 GB                            | Technical troubleshooting reference. Useful for net managers debugging FieldCommand.             |
| Project Gutenberg (public domain books) | ~65 GB (full)<br>/ ~1 GB (best-of) | Field reference library. Includes FCC regulations, ARRL handbook excerpts, first aid guides.     |

| CONTENT                                 | APPROX SIZE | BEST FOR                                                                                      |
|-----------------------------------------|-------------|-----------------------------------------------------------------------------------------------|
| Khan Academy (education)                | ~30 GB      | Training resource for shelter and EOC staff during extended activations.                      |
| Red Cross First Aid (WikiMed companion) | ~200 MB     | Practical first aid procedures in plain language — supplements WikiMed for non-medical staff. |
| Wiktionary (dictionary)                 | ~520 MB     | Reference for unfamiliar ICS terminology and acronyms.                                        |
| Amateur Radio Wiki (if available)       | Variable    | Band plans, antenna theory, Winlink and APRS procedures.                                      |

## How to Install Any ZIM File

```
# Step 1 – Find and download the ZIM from a device with internet
# Browse to: https://library.kiwix.org
# Filter by language (English) and category.
# Download the .zim file to your laptop or phone.

# Step 2 – Transfer the ZIM to the Pi
# Option A: USB drive (copy to USB, insert in Pi)
sudo mount /dev/sda1 /mnt
cp /mnt/my_content.zim /opt/kiwix/zim/

# Option B: SCP from your laptop over EMCOMM-NET
scp my_content.zim fieldcommand@192.168.50.1:/opt/kiwix/zim/

# Step 3 – Register the new ZIM with the Kiwix library index
sudo kiwix-manage /opt/kiwix/library.xml add /opt/kiwix/zim/my_content.zim

# Step 4 – Restart Kiwix to load the new content
sudo systemctl restart kiwix

# Step 5 – Verify it appears at http://192.168.50.1:8081
# The new content appears in the Kiwix home page search bar.

# To list all installed ZIM files:
ls -lh /opt/kiwix/zim/

# To remove a ZIM file:
sudo rm /opt/kiwix/zim/my_content.zim
sudo systemctl restart kiwix
```

ZIM files can be large. Check available disk space before downloading: `df -h /opt/kiwix/` The RAID array provides 2× 1 TB = ~900 GB usable after OS and FieldCommand. Tier 1 uses ~2.5 GB. Tier 2 uses ~10 GB. There is ample space for additional content.

## 7

## Printer Setup

## Step 7: Printer Setup

FieldCommand has print buttons on 17 pages. All printing uses the browser standard `window.print()` function, which sends the job to whatever printer the operator's browser can reach. Three printer connection methods are supported:

### Option A — Client Device Prints to Its Own Printer (Simplest)

Each operator's laptop or tablet prints to its own locally connected printer. No Pi configuration needed. Zero setup required — works immediately.



This is the simplest option and requires zero configuration on the Pi. The only requirement is that the operator's device has a printer configured.

### Option B — USB Printer Shared via Pi (CUPS)

A USB printer is connected directly to the Pi. CUPS shares it across EMCOMM-NET. Installed automatically by the FieldCommand installer. CUPS admin UI at <http://192.168.50.1:631>.

#### B.2 Add the USB Printer via CUPS Web Admin

| STEP | ACTION                                                                                                                         |
|------|--------------------------------------------------------------------------------------------------------------------------------|
| 1    | Plug the USB printer into one of the Pi's USB ports (or powered USB hub).                                                      |
| 2    | On any device on EMCOMM-NET, open a browser and go to: <b><a href="http://192.168.50.1:631">http://192.168.50.1:631</a></b>    |
| 3    | Click <b>Administration</b> → <b>Add Printer</b> .                                                                             |
| 4    | If prompted for credentials, enter the Pi username and password (e.g. fieldcommand / your password).                           |
| 5    | Your USB printer appears in the <b>Local Printers</b> list. Select it and click <b>Continue</b> .                              |
| 6    | Enter a Name (e.g. FieldCommand-Printer), Description, and Location. Check <b>Share This Printer</b> . Click <b>Continue</b> . |
| 7    | Select the correct printer driver. Click <b>Add Printer</b> .                                                                  |
| 8    | Set default options (paper size: Letter) and click <b>Set Default Options</b> .                                                |
| 9    | Click <b>Print Test Page</b> to confirm the printer is working.                                                                |

#### B.3 Add the Shared Printer on Operator Devices

| DEVICE TYPE                                | HOW TO ADD THE SHARED PRINTER                                                                                                                                                                                                         |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Windows laptop (Winlink / JS8Call station) | Settings → Bluetooth & devices → Printers & scanners → Add a printer or scanner. Windows discovers the CUPS shared printer automatically on EMCOMM-NET via Bonjour. Select FieldCommand-Printer when it appears and click Add device. |
| Mac / macOS                                | System Settings → Printers & Scanners → click + (Add Printer). Click the Default tab — the shared printer appears via Bonjour discovery. Select it and click Add.                                                                     |
| iPad / iPhone (AirPrint)                   | No setup required. CUPS + Avahi advertises the printer as AirPrint-compatible. Open any FieldCommand page → tap the Share icon → tap Print → select the printer. Works on any iOS device on EMCOMM-NET.                               |
| Android tablet / phone                     | Install the free CUPS Print app from Google Play. Open the app → Add Printer → enter IP 192.168.50.1, Port 631, Protocol IPP. The printer appears in the Android print dialog for all apps.                                           |

| DEVICE TYPE                                    | HOW TO ADD THE SHARED PRINTER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chromebook                                     | Settings → Advanced → Printing → Printers → Add Printer. Enter: Name = FieldCommand-Printer, Address = 192.168.50.1, Protocol = IPP, Queue = /printers/FieldCommand-Printer. Click Add.                                                                                                                                                                                                                                                                                                                                                                |
| Raspberry Pi 500 / 500+ (operator workstation) | Chromium browser (pre-installed): the CUPS printer appears automatically in the print dialog. No additional setup needed — Chromium on Pi OS discovers CUPS printers via Bonjour. For Firefox ESR on the Pi 500: install via <code>sudo apt install firefox-esr</code> then open Print → More Settings → select the printer from the destination list. Alternatively, run: <code>sudo usermod -aG lpadmin fieldcommand</code> to give the Pi 500 user full access to the CUPS admin at <a href="http://192.168.50.1:631">http://192.168.50.1:631</a> . |

## B.4 Recommended Printers for Field Use

### Monochrome Laser (recommended for most EOC deployments)

| PRINTER                       | CONNECTION   | WHY IT WORKS WELL                                                                                                                    |
|-------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Brother HL-L2350DW            | USB or Wi-Fi | Excellent Linux driver support out of the box. Fast, compact, duplex, durable. Best all-round choice for field EOC use. Under \$130. |
| HP LaserJet Pro M15w / P1102w | USB or Wi-Fi | Full Linux support via HP HPLIP driver (installed automatically with CUPS). Fast, reliable, very low cost per page. Under \$150.     |

### Color Multifunction Laser (ICS maps, colored forms, and chart printing)

| PRINTER                           | CONNECTION              | WHY IT WORKS WELL                                                                                                                                                   |
|-----------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Brother MFC-L3770CDW              | USB, Wi-Fi, or Ethernet | Color laser all-in-one with duplex, scan, copy, fax. Full Linux support via Brother driver. Prints color ICS maps and resource boards clearly. Under \$400.         |
| HP Color LaserJet Pro MFP M479fdw | USB, Wi-Fi, or Ethernet | Color laser multifunction — print, scan, copy, fax. HPLIP driver works with CUPS out of the box. Excellent for IAP packages with color-coded sections. Under \$500. |
| Canon imageCLASS MF743Cdw         | USB, Wi-Fi, or Ethernet | Color laser multifunction with excellent Linux CUPS driver (CAPT driver). Handles letter and legal paper. Good choice for permanent EOC installations. Under \$450. |

### Portable / Battery-Powered (mobile and field deployments without shore power)

| PRINTER           | CONNECTION                      | WHY IT WORKS WELL                                                                                                                                        |
|-------------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Canon PIXMA TR150 | USB or Wi-Fi (battery built-in) | Compact inkjet with rechargeable battery. Prints ~200 pages per charge. Letter size. Wi-Fi connects directly to EMCOMM-NET. Under \$200.                 |
| HP OfficeJet 200  | USB or Wi-Fi (battery optional) | Portable inkjet with larger paper tray than TR150. Optional battery accessory. Good for shelter check-in stations and mobile command posts. Under \$180. |

## Option C — Network Printer on EMCOMM-NET (Simplest Shared Setup)

A Wi-Fi or Ethernet printer is connected directly to the ASUS router or UniFi switch — no Pi involvement required at all. All devices on EMCOMM-NET can discover and print to it automatically via Bonjour / mDNS. This works with any network-capable printer including color multifunction laser printers that have built-in Wi-Fi or Ethernet. This is the simplest option for permanent EOC installations where a color MFP is already on site.

Option C works with any network-capable printer — including all three color multifunction laser printers listed in the BOM. These are the best choice for a permanent EOC installation because they connect directly to EMCOMM-NET with no Pi configuration, provide color printing for ICS maps and IAP packages, and are discoverable by every device on scene automatically.

| CONNECTION METHOD | COMPATIBLE PRINTERS                                                                                      | HOW TO SET UP                                                                                                                                                                                                                                             |
|-------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wi-Fi (wireless)  | Brother MFC-L3770CDW / HP Color LaserJet M479fdw / Canon MF743Cdw / Canon PIXMA TR150 / HP OfficeJet 200 | Use the printer control panel to join EMCOMM-NET. Enter the Wi-Fi password when prompted. The printer receives a 192.168.50.x address from the ASUS router automatically. All EMCOMM-NET devices discover it via Bonjour — no driver installation needed. |
| Ethernet (wired)  | Brother MFC-L3770CDW / HP Color LaserJet M479fdw / Canon MF743Cdw                                        | Run a CAT 6 cable from the printer Ethernet port to any spare UniFi Switch port (Ports 3 through 5). The printer receives an IP automatically. Set a static DHCP reservation in the ASUS router so the IP never changes between activations.              |

### Reserve a Static IP for the Printer (Recommended)

```
# On the ASUS router web UI (http://192.168.50.1):
# LAN → DHCP Server → Manually Assigned IP
# Enter the printer MAC address → assign a fixed IP, e.g.: 192.168.50.10
# Reboot the printer to pick up the reservation
# Confirm: ping 192.168.50.10
```

## Printer Option Comparison

|                               | Option A — Client Printer                  | Option B — USB via Pi / CUPS           | Option C — Network Printer             |
|-------------------------------|--------------------------------------------|----------------------------------------|----------------------------------------|
| Pi configuration needed       | None                                       | CUPS (auto-installed)                  | None                                   |
| One printer for all operators | Each device needs own printer              | Shared across EMCOMM-NET               | Shared across EMCOMM-NET               |
| Works without internet        | Yes                                        | Yes                                    | Yes                                    |
| iOS / AirPrint support        | Requires AirPrint printer                  | Yes via CUPS + Avahi                   | Yes if AirPrint printer                |
| Battery printer support       | Yes                                        | Yes (USB connected)                    | Yes (Wi-Fi connected)                  |
| Best for                      | Single operator or tablet with own printer | Any USB printer, shared to all devices | Wi-Fi/Ethernet printer already on site |

For most your callsign field activations, Option B (USB printer via CUPS) is recommended. A Brother HL-L2350DW or HP LaserJet plugged into the Pi USB hub covers all ICS forms, net logs, and cheat sheets needed for a typical activation.

## 8 First Boot Verification — Full System

### Step 8: First Boot Verification

After the installer completes and the Pi reboots, run through this verification checklist before handing off the system. Every service should show "active" and the dashboard should load cleanly from another device on EMCOMM-NET. If anything is not running, the troubleshooting section at the end of this guide covers the most common issues and their fixes.

```
# Check all FieldCommand services
for svc in fcc-lookup health-monitor deadmans ics-platform kiwix nginx pat; do
echo "$svc: $(systemctl is-active $svc)"
done

# Test the dashboard from the Pi itself
curl -s http://localhost/ | head -5
# Test API endpoints
curl http://localhost:5050/health # FCC API health check
curl http://localhost:5051/health # System health monitor
# Verify Pi has the correct static IP
ip addr show eth0
# Should show: inet 192.168.50.1/24
```

### Pre-Deployment Data Checklist

Before every activation, load the following data into FieldCommand. This can be done days ahead or immediately on-site — all data is stored locally and persists through reboots. The Preflight page (<http://192.168.50.1/preflight.html>) automatically verifies each of these items when it loads.

| DATA                 | HOW TO LOAD                                                                                                                                                                                                                                                                                     | WHERE TO VERIFY                                                                          | IMPACT IF MISSING                                                            |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Organization setup   | <a href="http://192.168.50.1/setup.html">http://192.168.50.1/setup.html</a> → enter org name, callsign, contact information, and deployment location.                                                                                                                                           | Preflight → Data Readiness → Organization Setup shows ✓                                  | ICS forms, cover pages, and header blocks will show blank fields             |
| FCC Amateur Database | Installed automatically by install.sh. Refresh: <code>sudo systemctl start fcc-refresh.service</code> (runs in background, ~10 minutes, ~600 MB download).                                                                                                                                      | Preflight → Data Readiness → FCC Callsign Database shows record count and age            | Net logger callsign lookup and AMPRNet authentication will not work          |
| Member Roster        | <a href="http://192.168.50.1/roster.html">http://192.168.50.1/roster.html</a> → Import CSV or + Add Member. CSV columns: first_name, last_name, callsign, radio_id, ema_id, city, state.                                                                                                        | Preflight → Data Readiness → Member Roster shows member count                            | Check-in forms will not auto-populate names; EMA ID lookup will not work     |
| Repeater Database    | <a href="http://192.168.50.1/repeaters.html">http://192.168.50.1/repeaters.html</a> → Offline File tab → log in at <a href="http://repeaterbook.com">repeaterbook.com</a> (free account) → search area → Export → CSV → drag CSV onto the drop zone. RepeaterBook CSV includes GPS coordinates. | Preflight → Data Readiness → Repeater Database shows count and how many have coordinates | ICS-205 channel picker will be empty; repeater map overlay will show no pins |

| DATA            | HOW TO LOAD                                                                                                                                                                                                                                                                     | WHERE TO VERIFY                                                  | IMPACT IF MISSING                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Channel Library | <a href="http://192.168.50.1/channel_library.html">http://192.168.50.1/channel_library.html</a> → + Add Channel, or click "+ Add to Channel Library" from the Repeater Database map popup.                                                                                      | Preflight → Data Readiness → Channel Library shows channel count | ICS-205 communications plan will have no channels to select                           |
| Active Incident | <a href="http://192.168.50.1/incident.html">http://192.168.50.1/incident.html</a> → + New Incident → enter incident name, type, location, and start time → Save. Or activate an event template at <a href="http://192.168.50.1/event_templates.html">event_templates.html</a> . | Preflight → Data Readiness → Active Incident shows ✓             | Net logger, resource board, ICS forms, and T-cards will have no incident to attach to |

The Preflight Deployment Checklist at <http://192.168.50.1/preflight.html> auto-checks all six items above on page load and shows live server detail — record counts, database age, and error descriptions — without any operator input. Run the Preflight page as the final go/no-go check before activating the system. Each NO-GO item includes a direct link to the page where it can be resolved.



## PART 2 — AMATEUR RADIO GROUP SETUP

**Skip this entire section if no licensed amateur radio group is involved in your deployment.** FieldCommand IMS operates fully and completely without any of the components in Part 2. All 49 web tools, ICS forms, net loggers, FEMA documentation, personnel accountability, and every other feature in Part 1 work independently of amateur radio hardware or services. Part 2 is for deployments where an ARES, RACES, amateur radio club, or other FCC-licensed amateur radio group is the lead or a key partner. A licensed amateur operator must lead the registration and setup for every item in this section.

| PART 2 SECTION                                 | WHAT IT INSTALLS / CONFIGURES                                                                                                                                                              | SKIP IF...                                                 |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| A — AMPRNet / 44Net Gateway                    | WireGuard tunnel to 44.0.0.0/8 amateur radio IP network. Dedicated gateway Pi. FCC callsign authentication.                                                                                | No licensed amateur group leads the deployment             |
| B — FCC Amateur Database                       | Offline SQLite database of 800K+ US amateur licensees. Enables callsign lookup in net loggers, check-in, and AMPRNet auth.                                                                 | No amateur radio net logging or callsign validation needed |
| C — APRS Setup (Direwolf + YAAC)               | APRS receive/transmit via TNC or radio interface. Direwolf software TNC + YAAC tactical display. Digirig, Signalink, or other USB TNC required.                                            | No APRS capability needed or no TNC available              |
| D — Pat Winlink                                | Winlink email-over-radio. Packet (VHF), VARA HF, VARA FM, Telnet. Callsign required. TNC or IC-7300 required for RF sessions.                                                              | No Winlink needed or no HF/VHF radio station               |
| E — Windows Laptop (Winlink Express + JS8Call) | Winlink Express + VARA HF on Windows. JS8Call HF keyboard messaging. IC-7300 or compatible HF transceiver required.                                                                        | No HF station or Windows laptop in the deployment          |
| F — SCS PACTOR Modem                           | PACTOR 1/2/3/4 HF digital modem for Winlink. Highest throughput and reliability on HF. SCS PXdragon DR-9400 (current flagship, USB-C + LAN) or P4dragon DR-7400 (USB-B). PACTOR 4 on both. | No PACTOR modem or PACTOR not required for your nets       |

### Part 2A — AMPRNet / 44Net Gateway Setup

#### What is AMPRNet and 44Net Connect?

AMPRNet — Amateur Packet Radio Network — is the global amateur radio IP network operating on the 44.0.0.0/8 address block, an entire Class A range permanently assigned by IANA to ARRL specifically for amateur radio use. Every licensed amateur radio operator in the world can apply for a block of 44.x.x.x IP addresses. These addresses are globally routable and reachable by any other AMPRNet participant, whether over RF (packet radio, APRS, mesh) or via an encrypted internet tunnel.

**44Net Connect** (connect.44net.cloud) is the current WireGuard-based tunneling service that makes AMPRNet accessible over the internet. The AMPRNet portal (portal.ampr.org) manages IP allocations and generates the WireGuard configuration for each registered operator. Once configured, the tunnel brings 44.0.0.0/8 directly to your gateway Pi — every device on EMCOMM-NET can reach any AMPRNet address worldwide as if it were a local IP address.

#### Is AMPRNet Right for Your Deployment?

**AMPRNet is beneficial when an amateur radio group is the lead organization or a key partner in the EMCOMM-NET deployment.** If FieldCommand is being deployed by or with an ARES, RACES, Amateur Radio Club, or other FCC-licensed amateur radio group, AMPRNet adds significant value: Winlink via radio-path gateways, APRS connectivity, inter-node FieldCommand communication, and access to a global amateur radio data network — all without touching the commercial internet.

If the deployment is primarily for public safety or served agency personnel without an amateur radio group involved, the AMPRNet gateway adds hardware cost and complexity with limited benefit. FieldCommand IMS runs entirely without AMPRNet — all 49 web pages, ICS forms, FEMA documentation, net loggers, personnel accountability, and all other tools operate locally on EMCOMM-NET with no dependency on AMPRNet. The 44Net Gateway Pi is an optional enhancement, not a requirement.

| SCENARIO                                           | AMPRNet RECOMMENDED?                      | REASON                                                                              |
|----------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------|
| ARES/RACES group leads the deployment              | ✓ Yes — strong benefit                    | Winlink via AMPRNet path, APRS-IS, inter-node data, global amateur station reach    |
| Amateur radio club operates the EOC comms          | ✓ Yes — strong benefit                    | Same as above — the licensed operators can register and maintain the allocation     |
| Amateur radio operators are supporting partners    | ■ Consider — moderate benefit             | Beneficial if the licensed operators take ownership of the AMPRNet registration     |
| Public safety agency only — no amateur radio group | ✗ Not recommended                         | Requires FCC amateur license for registration. Adds complexity with limited payoff. |
| Municipal EOC with some licensed staff             | ■ Consider — if a licensed operator leads | One designated licensed operator can register under their callsign and maintain it  |

## The Amateur Radio Group Must Lead This Process

The AMPRNet registration and maintenance process requires an FCC-licensed amateur radio operator. It cannot be done by a public safety agency, a municipality, or an IT department acting alone. The IP block is assigned to a callsign, maintained under that callsign, and all use is subject to Part 97 rules. The amateur radio group must own this — the served agency or EOC cannot.

In practical terms, this means the process should be initiated and led by the ARES Emergency Coordinator (EC), club President, or a designated technical lead who holds an FCC amateur license. The following organizational steps should happen before any hardware is purchased or any software is configured:

| STEP                                               | WHO DOES IT                      | WHAT HAPPENS                                                                                                                                                                                                                                                                           |
|----------------------------------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Designate the AMPRNet technical lead               | ARES EC or club President        | Identify a licensed amateur (any class) who will own the registration, maintain the allocation, and be the point of contact for AMPRNet administrators. Their callsign becomes the registered callsign for the subnet.                                                                 |
| Coordinate with the regional AMPRNet administrator | Designated technical lead        | AMPRNet is regionally administered. In Illinois, contact the Illinois AMPRNet coordinator (reachable via ARRL Illinois or the AMPRNet mailing list) to understand what address blocks are available in your area and to establish a relationship before submitting the formal request. |
| Agree on the subnet size and naming                | Technical lead + club leadership | A /29 gives 6 usable addresses (enough for one FieldCommand deployment plus growth). A /28 gives 14 addresses (suitable for multiple fixed and portable stations). The technical lead proposes this in the registration justification.                                                 |
| Establish a club or group callsign if needed       | Club leadership                  | The allocation can be under a personal callsign (e.g. KE4CON) or a club callsign (e.g. K9ESV). A club callsign is preferable for organizational deployments — it survives individual license changes and transfers more easily.                                                        |
| Plan for long-term maintenance                     | Club leadership + technical lead | AMPRNet allocations are permanent but require the license to remain active and the portal account to be maintained. Identify a backup technical lead in case the primary moves or is unavailable at activation time.                                                                   |

## What AMPRNet Enables — Capabilities and Restrictions

| CAPABILITY                            | DESCRIPTION                                                                                                                                                                                                                                  | REQUIRES WAN?                                                                                                      |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Winlink via AMPRNet path              | Connect to Winlink RMS gateways that have 44.x.x.x addresses. Keeps Winlink traffic within the amateur radio network rather than the commercial internet. Useful when you want to avoid commercial internet dependency for message handling. | Yes — WireGuard tunnel requires internet to establish, but once up, Winlink sessions travel over the 44-net tunnel |
| APRS-IS via AMPRNet                   | APRS-IS network servers are reachable on 44.x.x.x addresses. Configure Direwolf or YAAC to use the AMPRNet path instead of the public internet.                                                                                              | Yes — same tunnel dependency                                                                                       |
| Inter-node FieldCommand               | Two FieldCommand systems with 44Net gateways can share net log data, resource board status, and incident information over AMPRNet without routing through the commercial internet. Useful for multi-site EOC deployments.                    | Yes — tunnel required                                                                                              |
| Direct 44.x.x.x station access        | Any amateur station worldwide reachable on 44.x.x.x — whether via RF packet, mesh networking, or WireGuard — is directly addressable from any EMCOMM-NET device.                                                                             | Yes — tunnel required                                                                                              |
| Static globally routable IP addresses | Your 44.x.x.x block is permanently yours and globally routable. Other AMPRNet participants can reach your stations at a fixed address that never changes, regardless of your ISP or location.                                                | No — addresses are permanent; reachability requires tunnel                                                         |

**Part 97 restrictions apply to all AMPRNet traffic:** No encryption of content (WireGuard tunnel encryption of transport is permitted). No commercial traffic. Station identification required. No third-party traffic unless authorized. AMPRNet is a tool for the amateur radio community — public safety agencies using it through an amateur radio group must ensure all traffic complies with Part 97.

## How to Get Your AMPRNet IP Allocation — Overview

The process is straightforward but not fast. Allow **2–6 weeks** from initial registration to receiving a confirmed allocation. Regional administrators are volunteers and response times vary. Start this process well before your planned activation date.

| PHASE                     | ACTION                                                                                                                                                                                                                                                                                               | TIMELINE        |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1 — Coordinate locally    | Contact your regional AMPRNet coordinator before registering. Ask what blocks are available in your area, introduce your organization, and confirm they are aware of the incoming request. Illinois: check ARRL Illinois Section or the AMPRNet-Illinois mailing list.                               | 1–2 days        |
| 2 — Create portal account | Go to <a href="https://portal.ampr.org">https://portal.ampr.org</a> and register. Use your FCC callsign as your username. The portal verifies your license automatically against the FCC ULS database.                                                                                               | 1 day           |
| 3 — Request subnet        | Log in → Subnets → Request Subnet. Select your region. Request a /29 (6 addresses) or /28 (14 addresses). In the justification field, describe your organization, your role in EMCOMM, the number of fixed and portable stations, and intended use. A clear, detailed justification speeds approval. | 1 day to submit |
| 4 — Wait for approval     | The regional AMPRNet administrator reviews your request and may contact you with questions. Once approved, your allocation appears in <a href="https://portal.ampr.org">portal.ampr.org</a> → Subnets → My Subnets with status Active.                                                               | 1–6 weeks       |

| PHASE                         | ACTION                                                                                                                                                                                                                                                               | TIMELINE                 |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 5 — Download WireGuard config | After approval, the portal generates a WireGuard configuration specific to your subnet. Download it — it contains your private key, your 44.x.x.x address, and the connect.44net.cloud endpoint details. Store this securely — it is the credential for your tunnel. | Immediate after approval |
| 6 — Configure gateway Pi      | Follow Part B of this installation guide to configure the gateway Pi with the WireGuard config from the portal.                                                                                                                                                      | 1–2 hours                |

Keep a printed copy of your portal login, assigned subnet, and WireGuard configuration in your FieldCommand go-kit. If the gateway Pi needs to be rebuilt at a future activation, you will need these credentials. The portal generates a new WireGuard private key if needed — store the portal login credentials, not just the key file.

## AMPRNet in FieldCommand — What the Software Does

The following files are part of FieldCommand and work together to provide integrated 44Net gateway support across both Pis:

| FILE                            | RUNS ON                        | PURPOSE                                                                                                                                       |
|---------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| scripts/setup_44net.sh          | Gateway Pi (192.168.50.2)      | Installer for the gateway Pi. Sets up WireGuard, static IP, IP forwarding, status service, and Desktop autostart. Run once on the gateway Pi. |
| python/amprgate_status.py       | Gateway Pi (192.168.50.2)      | Flask status service on port 9000. Serves the gateway dashboard UI and /api/status JSON endpoint. Installed by setup_44net.sh.                |
| systemd/amprgate-status.service | Gateway Pi (192.168.50.2)      | Systemd unit that keeps amprgate_status.py running. Installed by setup_44net.sh. Starts after wg-quick@ampr0.                                 |
| python/amprgate_poll.py         | FieldCommand Pi (192.168.50.1) | Polls the gateway Pi every 30 seconds and writes status to /opt/fieldcommand/data/amprgate_status.json for the dashboard.                     |
| systemd/amprgate-poll.service   | FieldCommand Pi (192.168.50.1) | Systemd unit for the poll service. Installed automatically by the FieldCommand install.sh.                                                    |
| html/amprgate.html              | FieldCommand Pi (192.168.50.1) | Dashboard page at http://192.168.50.1/amprgate.html. Shows tunnel state, AMPRNet address, traffic stats, and tunnel controls.                 |

AMPRNet (Amateur Packet Radio Network) is the global amateur radio IP network using the reserved 44.0.0.0/8 address block assigned by IANA to ARRL. It provides licensed amateur radio operators with globally routable IP addresses reachable over RF links or via encrypted internet tunnels to the AMPRNet gateway. Adding a dedicated 44Net gateway Pi to FieldCommand gives every device on EMCOMM-NET full access to the AMPRNet — Winlink gateways, APRS-IS servers, Mesh networking nodes, and other amateur stations worldwide that are reachable on 44.x.x.x addresses.

The 44Net Gateway Pi is completely separate from the FieldCommand server Pi. It runs WireGuard, ip routing, and a status page only. If the gateway goes down or needs reconfiguring, FieldCommand continues operating normally. This isolation was an intentional design choice — keep routing infrastructure separate from the communications server.

## Part A — Get Your AMPRNet Allocation (Do This First — Allow 2-4 Weeks)

Before configuring any software, you need an AMPRNet IP address block assigned to your callsign. This is a one-time registration process managed by the AMPRNet administrators at ampr.org. The block you receive will be something like 44.x.x.x/29 (6 usable addresses) or 44.x.x.x/28 (14 usable addresses) depending on what is available in your region.

| STEP | ACTION                                                                                                                                                                                                                                                                              |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>Create an account at portal.ampr.org</b> — go to <a href="https://portal.ampr.org">https://portal.ampr.org</a> and click Register. You must use your callsign your callsign as your username. Verify your license via the FCC ULS lookup that the portal performs automatically. |

|   |                                                                                                                                                                                                                                                                                                                                                 |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | <b>Request a subnet allocation</b> — after logging in, click Subnets → Request Subnet. Select your region (Illinois — typically 44.24.x.x or 44.172.x.x range). Request a /29 block for a small deployment or a /28 if you anticipate growth. In the justification field, explain the your organization EMCOMM-NET deployment and intended use. |
| 3 | <b>Coordinate with your regional AMPRNet administrator</b> — Illinois AMPRNet administration is coordinated through ARRL. You may be contacted by the regional admin for verification. Response times vary from days to several weeks. Be patient.                                                                                              |
| 4 | <b>Receive your allocation</b> — you will get an email with your assigned 44.x.x.x/29 block. The portal will also show your allocation under Subnets → My Subnets. Record this allocation — you will need it in Part B.                                                                                                                         |
| 5 | <b>Note your WireGuard endpoint credentials</b> — the portal provides a WireGuard configuration for your specific subnet. Download or copy this — it contains your private key, allowed IPs, and the amprgw.ampr.org endpoint address.                                                                                                          |

ARRL membership is not required for an AMPRNet allocation, but a valid FCC amateur radio license (any class) is required. The your club or personal callsign is valid for this purpose. The allocation is tied to the callsign and is permanent as long as the license remains active.

## Part B — Set Up the 44Net Gateway Pi

Once you have your AMPRNet allocation, the following steps configure the dedicated gateway Pi running in the Argon NEO 5 case. This Pi should be freshly flashed with Raspberry Pi OS Lite (64-bit). It does not need the FieldCommand installer — it runs WireGuard and routing only.

### B.1 Initial OS Setup

Flash Raspberry Pi OS Desktop (64-bit) to the M.2 SATA SSD using Raspberry Pi Imager. Desktop is used on the gateway Pi so any team member can check tunnel status from the local keyboard and monitor without SSH. Chromium opens the status page automatically on login. During Imager advanced options set:

| OPTION     | VALUE                                                                                           |
|------------|-------------------------------------------------------------------------------------------------|
| Hostname   | amprgate.local — discoverable as amprgate.local on EMCOMM-NET                                   |
| Username   | amprgate — used by the gateway setup script                                                     |
| Password   | Choose a strong password and record it with the FieldCommand credentials                        |
| Enable SSH | Yes — useful for remote diagnostics if the monitor is not available                             |
| Wi-Fi      | Your home or lab Wi-Fi for initial package downloads — switched to wired EMCOMM-NET after setup |
| Locale     | US/Central — matches the FieldCommand Pi for consistent log timestamps                          |

### B.2 Assign Static IP and Install WireGuard

```
# SSH into the gateway Pi
ssh amprgate@[current-pi-ip]

# Update packages
sudo apt-get update && sudo apt-get full-upgrade -y

# Install WireGuard
sudo apt-get install -y wireguard wireguard-tools

# Set static IP on eth0
sudo nmcli con mod "Wired connection 1", ipv4.addresses 192.168.50.2/24, ipv4.gateway 192.168.50.254,
ipv4.dns "8.8.8.8 8.8.4.4", ipv4.method manual
sudo nmcli con up "Wired connection 1"

# Verify
ip addr show eth0
# Expected: inet 192.168.50.2/24
```

### B.3 Configure the WireGuard Tunnel

The AMPRNet portal provides a WireGuard configuration specific to your subnet. The following is the general structure — replace the placeholder values with the actual values from your portal allocation.

```
# Create the WireGuard config directory
sudo mkdir -p /etc/wireguard
sudo chmod 700 /etc/wireguard

# Create the tunnel config file
sudo nano /etc/wireguard/ampr0.conf

# Paste the following – replace ALL values in [brackets] with your portal values:
[Interface]
PrivateKey = [your-private-key-from-portal]
Address = [your-44.x.x.x/29-address]
ListenPort = 51820
PostUp = ip route add 44.0.0.0/8 dev ampr0
PostUp = iptables -A FORWARD -i ampr0 -j ACCEPT
PostUp = iptables -A FORWARD -o ampr0 -j ACCEPT
PostUp = iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
PostDown = ip route del 44.0.0.0/8 dev ampr0
PostDown = iptables -D FORWARD -i ampr0 -j ACCEPT
PostDown = iptables -D FORWARD -o ampr0 -j ACCEPT
PostDown = iptables -t nat -D POSTROUTING -o eth0 -j MASQUERADE

[Peer]
PublicKey = [amprnet-gateway-public-key-from-portal]
Endpoint = amprgw.ampr.org:51820
AllowedIPs = 44.0.0.0/8
PersistentKeepalive = 25
```

### B.4 Enable IP Forwarding and Start the Tunnel

```
# Enable IP forwarding (required for routing between EMCOMM-NET and AMPRNet)
echo "net.ipv4.ip_forward = 1" | sudo tee -a /etc/sysctl.conf
sudo sysctl -p

# Start the WireGuard tunnel
sudo wg-quick up ampr0

# Enable at boot
sudo systemctl enable wg-quick@ampr0

# Check tunnel status
sudo wg show ampr0
# Expected: shows latest handshake, transfer stats, allowed IPs

# Test reachability into AMPRNet
ping 44.0.0.1 -c 4
# Expected: replies from 44.0.0.1 (AMPRNet gateway)
```

### B.5 Access Control — Who Can Reach What

FieldCommand restricts access to the 44Net gateway on two levels. This is both a security measure and a Part 97 compliance measure — only licensed amateur radio operators should be able to view or control the AMPRNet gateway.

| PORT                  | ACCESSIBLE FROM                                                   | AUTH REQUIRED                                                            | PROVIDES                                                     |
|-----------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------|
| Port 9000 (0.0.0.0)   | Any EMCOMM-NET device (192.168.50.x)                              | None for /api/status (machine poll). Valid FCC callsign for web UI.      | Read-only status JSON. Callsign-authenticated web dashboard. |
| Port 9001 (127.0.0.1) | Gateway Pi keyboard only (localhost — not reachable from network) | Valid callsign session from port 9000 login. Physical presence required. | Tunnel control: up / down / restart.                         |

Tunnel controls (up/down/restart) are intentionally inaccessible from the network. An operator who wants to restart the WireGuard tunnel must sit at the gateway Pi keyboard, open Chromium, log in with their FCC callsign, and use the local control interface. This prevents accidental or unauthorized tunnel disruption from operator laptops or phones. The FieldCommand Pi polling service reads /api/status on port 9000 without authentication — this is a read-only machine-to-machine call and is intentional.

### B.5 Advertise the 44Net Route to EMCOMM-NET Devices

For other devices on EMCOMM-NET to use the 44Net gateway automatically, the ASUS router needs to push a static route to all DHCP clients. This tells every device: "to reach anything in 44.0.0.0/8, use 192.168.50.2".

| STEP | ACTION                                                                                                                                                |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Open the ASUS router admin at <b>http://192.168.50.254</b> .                                                                                          |
| 2    | Go to <b>LAN → Route</b> .                                                                                                                            |
| 3    | Click <b>Add</b> and enter: Network/Host IP = <b>44.0.0.0</b> , Netmask = <b>255.0.0.0</b> , Gateway = <b>192.168.50.2</b> , Interface = <b>LAN</b> . |
| 4    | Click <b>Apply</b> .                                                                                                                                  |
| 5    | On any EMCOMM-NET device, verify the route was received:                                                                                              |

```
# On any EMCOMM-NET device (Pi 500, Windows laptop, etc.)
ip route show | grep 44
# Expected: 44.0.0.0/8 via 192.168.50.2 dev eth0

# Windows equivalent (Command Prompt)
route print | find "44."
# Expected: 44.0.0.0 0.0.0.0 192.168.50.2 ...

# Test from the FieldCommand Pi
ping 44.0.0.1 -c 4
# If reachable, FieldCommand can now connect to AMPRNet resources
```

## B.6 Gateway Status Page

Install a lightweight status page on the 44Net Pi so operators can verify the tunnel state from any EMCOMM-NET browser without SSH access. This is a small Flask application that displays tunnel status, connected peers, and recent handshake times.

```

# Install Python and Flask on the 44Net Pi
sudo apt-get install -y python3 python3-pip python3-venv
python3 -m venv /opt/amprgate/venv
/opt/amprgate/venv/bin/pip install flask

# Create the status app
sudo mkdir -p /opt/amprgate
sudo nano /opt/amprgate/status.py

# Paste the following into status.py:
# ---
from flask import Flask, jsonify
import subprocess, json
app = Flask(__name__)

@app.route("/")
def index():
    return open("/opt/amprgate/status.html").read()

@app.route("/api/status")
def status():
    wg = subprocess.run(["sudo", "wg", "show", "ampr0", "dump"],
        capture_output=True, text=True).stdout
    lines = [l.split(" ") for l in wg.strip().split(" ") if l]
    return jsonify({"tunnel": "up" if lines else "down", "peers": len(lines)-1})

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=9000)
# ---

# Create systemd service
sudo nano /etc/systemd/system/amprgate-status.service
# Paste:
[Unit]
Description=AMPRNet Gateway Status Page
After=network.target wg-quick@ampr0.service

[Service]
User=amprgate
ExecStart=/opt/amprgate/venv/bin/python /opt/amprgate/status.py
Restart=always

[Install]
WantedBy=multi-user.target

sudo systemctl enable --now amprgate-status
# Status page now at: http://192.168.50.2:9000

```

## Part C — What You Can Do on AMPRNet

Once the gateway is operational and the route is advertised to EMCOMM-NET, every device on EMCOMM-NET has full access to the AMPRNet. Here are the most useful applications for your organization operations:

| HOW IT USES 44NET                                                                                                                                                                                                                    | ACCESS FROM EMCOMM-NET                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Some Winlink RMS gateways have 44.x.x.x addresses. Pat Winlink on the FieldCommand Pi can connect to them directly over the AMPRNet tunnel rather than the commercial internet — useful when only amateur radio paths are available. | Pat Winlink at <a href="http://192.168.50.1:8090">http://192.168.50.1:8090</a> Add AMPRNet RMS |
| The APRS-IS network has servers reachable on 44.x.x.x. Direwolf and YAAC can connect to APRS-IS via the AMPRNet path, keeping APRS traffic within the amateur radio network.                                                         | Configure APRS-IS server address in Direwolf or YAAC set                                       |
| If another your organization station runs a second FieldCommand system with its own 44Net gateway, the two systems can share net data and resource boards over AMPRNet without routing traffic through the commercial internet.      | Future capability — requires second FieldCommand deploy                                        |
| Some VoIP amateur radio nodes are reachable on 44.x.x.x addresses, providing RF-to-AMPRNet linking without commercial internet dependency.                                                                                           | Direct connection to 44.x.x.x node addresses                                                   |
| Any amateur station worldwide with a 44.x.x.x address (whether via RF packet or WireGuard tunnel) is directly reachable from any EMCOMM-NET device for data exchange, file transfer, or status checking.                             | Direct TCP/IP connections to 44.x.x.x addresses                                                |

AMPRNet does not carry voice traffic and is not a replacement for your radio links. It is a data network for IP-based amateur radio applications. All traffic on AMPRNet is subject to Part 97 rules — no encryption of content, no commercial traffic, licensed operator identification required. WireGuard encryption of the tunnel itself is permitted as it is the transport, not the content.

## Part D — Verification Checklist

| CHECK                                     | COMMAND / METHOD                                                                                  | EXPECTED RESULT                                                         |
|-------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| AMPRNet portal shows allocation           | Login to <a href="http://portal.ampr.org">portal.ampr.org</a> → Subnets → My Subnets              | Your 44.x.x.x/29 block shows Status: Active                             |
| WireGuard tunnel is up on gateway Pi      | SSH to 192.168.50.2 then:<br>sudo wg show ampr0                                                   | Shows latest handshake within last 2 minutes                            |
| AMPRNet gateway reachable from gateway Pi | ping 44.0.0.1 -c 4                                                                                | Replies received — round trip time typically 10-50 ms                   |
| EMCOMM-NET devices have 44 route          | ip route show   grep 44 (on any EMCOMM-NET device)                                                | 44.0.0.0/8 via 192.168.50.2                                             |
| AMPRNet reachable from FieldCommand Pi    | ping 44.0.0.1 -c 4 (on FieldCommand Pi)                                                           | Replies received                                                        |
| Status page accessible                    | Open <a href="http://192.168.50.2:9000">http://192.168.50.2:9000</a> in any browser on EMCOMM-NET | Shows callsign login page — enter a valid FCC callsign to proceed       |
| Callsign login works                      | Enter your callsign at <a href="http://192.168.50.2:9000">http://192.168.50.2:9000</a> login page | FCC database validates callsign — dashboard loads showing tunnel status |
| Access log recording                      | SSH to gateway Pi: tail /var/log/amprgate-access.log                                              | Shows LOGIN-OK entry with your callsign, IP, and license class          |

| CHECK                                     | COMMAND / METHOD                                           | EXPECTED RESULT                                  |
|-------------------------------------------|------------------------------------------------------------|--------------------------------------------------|
| Tunnel control blocked from network       | From a laptop on EMCOMM-NET: curl http://192.168.50.2:9001 | Connection refused — port 9001 is localhost-only |
| AMPRNet reachable from Pi 500 workstation | Open http://192.168.50.2:9000 in Chromium on Pi 500        | Status page loads and shows tunnel UP            |

## B FCC Amateur Database (Part 2B)

### Part 2B — FCC Amateur Database

The FCC database gives FieldCommand instant offline access to the entire US amateur radio license database — approximately 800,000 active licensees. When a net control operator types a callsign into the check-in field, FieldCommand looks up the licensee name, license class, and address automatically in under a millisecond, with no internet connection required. This is one of the most-used features during ARES/RACES net operations.

The installer offers to download and build the database during Step 4 (answer Y). If you skipped it during installation, or if the database needs to be rebuilt, run the commands below. Building takes approximately 5 to 10 minutes on a Pi 5 and requires an active internet connection.

```
# If you selected "y" during install, it already ran.
# To build or rebuild manually:
sudo -u fieldcommand \
/opt/fieldcommand/venv/bin/python \
/opt/fieldcommand/python/build_fcc_db.py
# Automatic weekly refresh timer:
sudo systemctl status fcc-refresh.timer
# Force immediate refresh:
sudo systemctl start fcc-refresh.service
```

## C

## APRS Setup — Direwolf TNC + YAAC (Part 2C)

## Part 2C — APRS Setup (Direwolf TNC + YAAC)

Direwolf is the software TNC that decodes APRS off the radio. It exposes a KISS interface (TCP port 8001) and an AGWPE interface (TCP port 8000) — but it has NO web or REST interface of its own. Live stations reach the Tactical APRS Map from APRS Command (on the Windows laptop), which connects to Direwolf and serves the map its station feed; a small Pi-side KISS bridge can serve the same feed if you want the map to work without the laptop. YAAC (port 8082) is an optional on-Pi APRS client.

APRS is entirely optional. FieldCommand runs fully without it — you simply will not see live APRS stations on the tactical map. If you are not planning to use a TNC or radio interface, skip this step entirely and proceed to Step 8 (Printer Setup).

The FieldCommand installer installs Direwolf (via apt) and YAAC automatically, writes a starter `/etc/direwolf.conf`, and enables `direwolf.service`. If the installer succeeded, you only need to edit the config (Section 9.3).

### 9.1 Verify Automatic Installation

```
# Direwolf — software TNC (no jar; installed via apt)
which direwolf && direwolf -h | head -1
sudo systemctl status direwolf
# YAAC — optional Java client
java -version
ls -lh /opt/yaac/YAAC.jar
sudo systemctl status yaac
```

### 9.2 Manual Install — If Automatic Install Failed

```
# Direwolf — software TNC
sudo apt install -y direwolf
sudo systemctl enable --now direwolf
# YAAC (optional) — needs Java
sudo apt install -y default-jre
sudo mkdir -p /opt/yaac
cd /tmp && wget http://www.ka2ddo.org/ka2ddo/YAAC.zip
sudo unzip YAAC.zip "*.jar" -d /opt/yaac/
sudo mv /opt/yaac/YAAC*.jar /opt/yaac/YAAC.jar 2>/dev/null || true
sudo chown -R fieldcommand:fieldcommand /opt/yaac
sudo systemctl enable --now yaac
```

### 9.3 Configure Direwolf (Required — First Run)

| STEP | ACTION                                                                                                                                      |
|------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Find your audio interface: <code>arecord -l</code> (note the card,device numbers).                                                          |
| 2    | Edit <code>sudo nano /etc/direwolf.conf</code> : set <b>ADEVICE</b> (e.g. <code>plughw:1,0</code> ) and <b>MYCALL</b> (your callsign-SSID). |
| 3    | Restart: <code>sudo systemctl restart direwolf</code> .                                                                                     |
| 4    | Direwolf is receive-only by default. To transmit (iGate / digipeat), configure PTT in the config (GPIO / CM108 / rig control).              |

### 9.4 Feed the Map — APRS Command or a KISS Bridge

Direwolf itself does not serve the map. Point the map (Tactical → Settings → APRS Sources) at whatever reads Direwolf's KISS/AGW output and serves stations:

| SOURCE                         | CONNECTS TO DIREWOLF VIA                    | MAP SETTING                         |
|--------------------------------|---------------------------------------------|-------------------------------------|
| APRS Command (Windows laptop)  | KISS TCP :8001 or AGW :8000 over EMCOMM-NET | RF source host = laptop IP          |
| Pi-side KISS bridge (optional) | KISS TCP localhost:8001                     | RF source host = localhost          |
| APRS-IS internet (if WAN up)   | internet APRS-IS                            | enable the APRS-IS layer on the map |

## 9.5 Verify APRS is Working

```
# Watch Direwolf decode frames off the radio:
journalctl -u direwolf -f
# Confirm the KISS/AGW ports are listening (for APRS Command):
ss -tlnp | grep -E "8000|8001"
# Stations appear on the Tactical Map once APRS Command (or the bridge) is running.
```

## Part 2D — Pat Winlink — Verify & Configure

Pat is an open-source Winlink client that runs directly on the Pi and is accessible from any browser on EMCOMM-NET at <http://192.168.50.1:8090>. The FieldCommand installer downloads and configures Pat automatically, so this step is primarily about verifying it is running and adding your Winlink account password.

Pat serves as the backup Winlink path when the Windows laptop is unavailable, or for VHF Packet operations when a TNC is connected directly to the Pi. your organization primary Winlink operations use Winlink Express on the Windows laptop (Step 10). Pat is the fallback.

```
# Check pat.service status
sudo systemctl status pat
# If not running:
sudo systemctl start pat && sudo systemctl enable pat
# Confirm port 8090 is listening
ss -tlnp | grep 8090
# Add your Winlink password:
sudo nano /opt/fieldcommand/.config/pat/config.json
# Verify / update these fields:
# "mycall": "your callsign" ← set by installer
# "secure_login_password": "xxxxx" ← add your Winlink password
# "http_addr": "0.0.0.0:8090" ← must be 0.0.0.0 for EMCOMM-NET access
```

## Part 2E — Windows Laptop Setup (Winlink Express + JS8Call)

The Windows laptop is the primary HF digital station. It runs Winlink Express and JS8Call with the IC-7300 connected via a single USB cable. Connect the laptop to EMCOMM-NET (Wi-Fi) or the UniFi switch Port 3 (wired Ethernet).

### IC-7300 One-Time Radio Setup

| IC-7300 MENU PATH                             | SETTING |
|-----------------------------------------------|---------|
| SET → Connectors → CI-V Baud Rate             | 115200  |
| SET → Connectors → CI-V Transceive            | ON      |
| SET → Connectors → USB Send/Keying → USB Send | RTS     |
| SET → Connectors → MOD Input → USB MOD Level  | 40–50%  |
| COMP button (speech compression)              | OFF     |
| Mode for digital operation                    | USB-D   |

### Install Winlink Express + VARA HF

| STEP | ACTION                                                                                                                                              |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Download Winlink Express from: <a href="http://winlink.org/client-software">winlink.org/client-software</a> . Run installer, accept defaults.       |
| 2    | On first launch: enter your callsign, Winlink password, and grid square (your county).                                                              |
| 3    | Settings → Radio Setup: Radio=IC-7300, Control Port=(check Device Manager), Baud Rate=115200, PTT via CAT=checked.                                  |
| 4    | Settings → Sound Card: Input=USB Audio CODEC (IC-7300), Output=USB Audio CODEC (IC-7300).                                                           |
| 5    | Download and install VARA HF from: <a href="http://winlink.org">winlink.org</a> → VARA → VARA HF Modem. Set same audio devices in VARA HF Settings. |
| 6    | Test: Open a VARA HF session in Winlink Express → connect to any Winlink RMS gateway on 40m to confirm audio and CAT control work.                  |

### Install JS8Call

| STEP | ACTION                                                                                                                                                                                 |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Download JS8Call from: <a href="http://js8call.com">js8call.com</a> → Windows installer. Run installer, accept defaults.                                                               |
| 2    | Launch JS8Call. Open File → Settings (F2). General tab: My Call=, My Grid=.                                                                                                            |
| 3    | Audio tab: Input=USB Audio CODEC (IC-7300), Output=USB Audio CODEC (IC-7300).                                                                                                          |
| 4    | Radio tab: Rig=IC-7300, PTT Method=CAT, Serial Port=(same COM port as Winlink), Baud Rate=115200.                                                                                      |
| 5    | Reporting tab (CRITICAL): TCP Server Hostname= <b>0.0.0.0</b> (MUST change from 127.0.0.1). Enable TCP Server API=checked. TCP Server Port= <b>2442</b> . Accept TCP Requests=checked. |
| 6    | Click OK. Restart JS8Call for API settings to take effect.                                                                                                                             |
| 7    | Set VFO to 7.078 MHz (JS8 40m calling frequency). Mode → Normal.                                                                                                                       |

### Windows Firewall — Allow JS8Call Port 2442

1. Search "Windows Defender Firewall" → Advanced Settings
2. Inbound Rules → New Rule → Port → TCP → Specific port: 2442 → Allow → All profiles
3. Name the rule: JS8Call API

## Find the Windows Laptop IP on EMCOMM-NET

```
# Run in Windows Command Prompt:
ipconfig
# Note the IPv4 Address for the EMCOMM-NET adapter (e.g. 192.168.50.105)
# Set a DHCP reservation (recommended – gives laptop a fixed IP):
# ASUS Web UI → LAN → DHCP Server → Manually Assigned IP
# Enter laptop MAC address → assign 192.168.50.2

# Configure the FieldCommand dashboard JS8Call card:
# 1. Connect to EMCOMM-NET, open http://192.168.50.1
# 2. Find the JS8Call card (purple) in Amateur Radio section
# 3. Tap/click the card
# 4. Enter the Windows laptop IP (e.g. 192.168.50.105)
# 5. Card saves IP and opens http://192.168.50.105:2442
```

| SWITCHING TO...                  | PROCEDURE                                                                                                                                                                           |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Winlink Express (from JS8Call)   | Click Disconnect in JS8Call or close it. Open VARA HF — it automatically reclaims the USB audio device. Open Winlink Express and connect normally.                                  |
| JS8Call (from Winlink)           | Finish any active Winlink transmission. Close VARA HF. Minimize Winlink Express. Open JS8Call and click Connect — it reclaims the COM port and audio.                               |
| Both simultaneously (two radios) | Connect a second HF radio via a separate USB audio and CAT interface. Configure JS8Call to use that radio's COM port and audio device. Both run at the same time without conflicts. |

## Part 2F — SCS PACTOR Modem Setup

PACTOR is a high-performance HF digital mode used extensively in Winlink operations. It provides significantly better throughput than VARA HF on marginal or noisy HF paths, and is the only mode that supports PACTOR 4 compression on Winlink. SCS (Special Communications Systems) manufactures the only PACTOR-licensed TNCs available. A PACTOR TNC connects between the IC-7300 and the Windows laptop. This step is optional — Winlink via VARA HF works well for most activations.

### Supported SCS PACTOR Modems

| MODEL                                   | PACTOR LEVEL         | INTERFACE                           | NOTES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------|----------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCS Tracker DSP TNC                     | PACTOR 1 / 2 / 3     | USB-B (COM port)                    | Entry-level PACTOR TNC. Good for most Winlink operations. Also supports PACKET and ROBUST PR. ~\$250.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| SCS PTC-IIIusb                          | PACTOR 1 / 2 / 3     | USB-B (COM port)                    | Full-featured controller with front-panel display. Supports PACTOR, PACKET, AMTOR, RTTY. ~\$650.                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| SCS P4dragon DR-7400                    | PACTOR 1 / 2 / 3 / 4 | USB-B (COM port)                    | Compact PACTOR 4 modem — proven platform. Good entry point to PACTOR 4. ~\$1,100. Software compatible with PTC-III template in Winlink Express.                                                                                                                                                                                                                                                                                                                                                                                                          |
| SCS PXdragon DR-9400 (current flagship) | PACTOR 1 / 2 / 3 / 4 | USB-C + LAN + Wi-Fi/BT (BTWF model) | Successor to the P4dragon series. Quad-core ARM v8 running real-time Linux. Up to 10,500 bps on PACTOR-4. Also supports Packet Radio (1k2 AFSK / 9k6 G3RUH), Robust Packet Radio, APRS beaconing, weather-fax (RX), RTTY (RX). GPS input (galvanically isolated). 1-ppm TCXO. 10 bi-color status LEDs. DR-9400 = standard; DR-9400-BTWF = includes Wi-Fi/Bluetooth dongle + license key. Software compatible with PTC-III/P4dragon apps: Winlink Express, Airmail, SafecomLink. Available at DX Engineering, WiMo, R&L; Electronics. ~\$1,400-1,600 USD. |

### Physical Connection

| FROM                        | TO                                                      | CABLE / NOTE                                             |
|-----------------------------|---------------------------------------------------------|----------------------------------------------------------|
| SCS TNC — AFSK audio output | IC-7300 — ACC jack audio input or 3.5mm mic input       | SCS-supplied audio cable (included with TNC)             |
| SCS TNC — AFSK audio input  | IC-7300 — ACC jack audio output or 3.5mm speaker output | SCS-supplied audio cable (included with TNC)             |
| SCS TNC — PTT output        | IC-7300 — remote jack (ACC or SEND)                     | SCS-supplied PTT cable (check your TNC model)            |
| SCS TNC — USB port          | Windows laptop — any USB-A port                         | USB-A to USB-B or USB-A to micro-USB (included with TNC) |

If you are also using VARA HF via the IC-7300 USB audio: configure the PACTOR TNC to use the IC-7300 ACC jack for audio, not the USB sound card. This allows VARA (USB audio path) and PACTOR (ACC audio path) to coexist on the same radio. Never transmit both modes simultaneously.

**PXdragon DR-9400 — LAN connection option:** The DR-9400 includes a LAN (Ethernet) port in addition to USB-C. When connected via LAN, Winlink Express reaches the modem as a TCP device: Settings → Radio Setup → Connection type → Network/TCP. Enter the DR-9400 LAN IP (check your router DHCP table) and port 8000. This lets the laptop connect to the modem over EMCOMM-NET without a direct USB cable — useful when the laptop and modem are at different positions in the EOC. The DR-9400-BTWF model also supports Wi-Fi and Bluetooth connections.

## Configure Winlink Express for PACTOR

| STEP | ACTION                                                                                                                                                                                       |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | In Winlink Express, go to <b>Settings</b> → <b>Radio Setup</b> .                                                                                                                             |
| 2    | Click <b>Add</b> to create a new radio entry.                                                                                                                                                |
| 3    | Set <b>Template</b> to: <b>SCS PTC-IIx</b> (works for all SCS PACTOR TNCs: Tracker, PTC-IIusb, P4dragon DR-7400, and PXdragon DR-9400 are all compatible with this template).                |
| 4    | Set <b>Control Port</b> to the COM port assigned to your SCS TNC. Check Windows Device Manager → Ports (COM & LPT) to find the correct number. The TNC appears as "USB Serial Port (COM X)". |
| 5    | Set <b>Baud Rate</b> to <b>115200</b> .                                                                                                                                                      |
| 6    | Leave <b>PTT via CAT</b> unchecked — the TNC handles PTT directly via the ACC cable.                                                                                                         |
| 7    | Click <b>OK</b> to save the radio profile.                                                                                                                                                   |

## Open a PACTOR Winlink Session

| STEP | ACTION                                                                                                                                                      |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | In Winlink Express, click the session type dropdown and select <b>Open Session</b> → <b>PACTOR Winlink</b> .                                                |
| 2    | The session window opens. The TNC initializes and you will see the SCS banner text in the session log.                                                      |
| 3    | Set your frequency on the IC-7300 to a known Winlink RMS gateway frequency for your region. Common 40m PACTOR frequencies: 7.038 MHz, 7.040 MHz, 7.103 MHz. |
| 4    | Click <b>Start</b> . Winlink Express scans for and connects to the nearest RMS gateway.                                                                     |
| 5    | The PACTOR handshake completes automatically — the session connects at the highest common level (PACTOR 1 through 4) that both stations support.            |
| 6    | Send and receive messages as normal. PACTOR 4 typically completes a full ICS-213 exchange in under 2 minutes on a good path.                                |

## Verify the TNC is Responding

```
# USB-B modems (Tracker, PTC-IIIusb, DR-7400):
# PuTTY → Connection type: Serial → COM port: (check Device Manager) → Speed: 115200

# PXdragon DR-9400 via USB-C:
# Same as above – DR-9400 appears as a COM port in Device Manager

# PXdragon DR-9400 via LAN:
# PuTTY → Connection type: Raw → Host: [DR-9400 LAN IP] → Port: 8000 → Open

# Type the following TNC command:
cmd
# The TNC responds with its command prompt: cmd:

# Useful TNC verification commands:
ver # shows firmware version and TNC model
status # shows current operating status
mycall YOURCALL # replace YOURCALL with your FCC callsign

# If the TNC does not respond:
# 1. Confirm the correct COM port in Device Manager
# 2. Confirm no other application has the COM port open
# 3. Try baud rate 9600 (some TNC models default lower)
```

PACTOR levels 3 and 4 require a separate Winlink license fee beyond the standard Winlink account. PACTOR 1 and 2 are available to all licensed amateur radio operators at no additional cost. The PXdragon DR-9400 and P4dragon series also support AX.25 Packet and Robust Packet Radio (RPR), accessible via KISS TCP or the DR-9400 LAN port. The DR-9400-BTWF model adds Wi-Fi/Bluetooth, enabling phone or tablet Winlink access without a laptop for portable deployments. These provide a VHF backup to the primary HF PACTOR path.

## Network Architecture Reference

### Full Network Topology

The EMCOMM-NET deployment uses the ASUS RT-BE58 Go as the Wi-Fi access point and DHCP server. The primary cellular modem provides WAN connectivity via a PoE Ethernet cable directly to the ASUS WAN port. satellite dish provides automatic secondary WAN failover via the ASUS USB WAN port. The UniFi Switch Lite 16 PoE distributes wired connections to both Pi servers, all operator workstations, the network printer, and additional field devices. The 44Net Gateway Pi routes the 44.0.0.0/8 AMPRNet address block for all EMCOMM-NET devices via a WireGuard tunnel to amprgw.ampr.org.

| DEVICE                                   | IP ADDRESS                              | ROLE                                                               | CONNECTION                                                                                          |
|------------------------------------------|-----------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| ASUS RT-BE58 Go                          | 192.168.50.254<br>(LAN gateway)         | Wi-Fi 7 AP + DHCP server<br>+ WAN gateway                          | WAN: site Ethernet, satellite dish, or USB tether (optional). EMCOMM-NET SSID on 2.4 GHz and 5 GHz. |
| UniFi Switch Lite 16 PoE                 | N/A (Layer 2 switch)                    | Wired distribution hub — 16 ports                                  | Port 1 uplink to ASUS router. Ports 2-16 to all wired devices.                                      |
| FieldCommand Pi 5<br>(Pironman MAX 5)    | 192.168.50.1<br>(static)                | FieldCommand application server All EmComm tools and services      | Wired GbE via UniFi Switch Port 2. Does not run Wi-Fi.                                              |
| 44Net Gateway Pi 5<br>(Argon NEO 5)      | 192.168.50.2<br>(static)                | AMPRNet / 44Net WireGuard gateway Routes 44.0.0.0/8 for EMCOMM-NET | Wired GbE via UniFi Switch Port 3. WireGuard tunnel to amprgw.ampr.org.                             |
| Windows Laptop                           | 192.168.50.3<br>(DHCP reservation)      | Winlink Express + JS8Call IC-7300 HF station                       | Wired via Port 4 or Wi-Fi (EMCOMM-NET).                                                             |
| Color MFP Printer                        | 192.168.50.10<br>(DHCP reservation)     | Network printer shared to all devices                              | Wired via Port 5 or Wi-Fi (EMCOMM-NET).                                                             |
| Pi 500 Workstations x4                   | 192.168.50.20-23<br>(DHCP reservations) | Operator browser stations                                          | Wired via Ports 6-9 or Wi-Fi (EMCOMM-NET).                                                          |
| ASUS RT-BE58 Go Mesh Node 1              | Managed by AiMesh                       | EMCOMM-NET extension (same SSID, seamless roaming)                 | Wired backhaul via UniFi Port 11. Extends coverage to secondary zone.                               |
| ASUS RT-BE58 Go Mesh Node 2              | Managed by AiMesh                       | EMCOMM-NET extension (same SSID, seamless roaming)                 | Wired backhaul via UniFi Port 12. Extends coverage to third zone.                                   |
| Field Devices (phones, tablets, laptops) | 192.168.50.100-200 (DHCP)               | Additional operator browsers                                       | Wi-Fi — EMCOMM-NET — roam between primary and mesh nodes automatically.                             |



Recommended: set the ASUS router LAN IP to 192.168.50.254 and the Pi static IP to 192.168.50.1. This keeps the FieldCommand URL clean and the router admin clearly separate at <http://192.168.50.254>.

## Web Dashboard Reference

FieldCommand IMS serves 49 web pages from the Pi. All are accessible at [http://192.168.50.1/\[page\]](http://192.168.50.1/[page]). Pages marked WAN require internet for live data but load and display cached data when offline.

| PAGE                        | URL                   | NOTES                                                                   |
|-----------------------------|-----------------------|-------------------------------------------------------------------------|
| Dashboard (All-Hazards ICS) | /                     | Main hub — 3 modes, WAN status bar, tool cards. Default entry point.    |
| Organization Setup          | /setup.html           | Configure org name, callsign, grid square, time zone, SSID              |
| Preflight Check             | /preflight.html       | GO/CAUTION/NO-GO readiness. Run before every activation.                |
| Incident Management         | /incident.html        | Create/manage incidents, operational periods, incident type             |
| Incident Archive/Reset      | /incident_mgmt.html   | Archive to USB · Restore · Hard delete · Beta/Scenario Reset            |
| Event Templates             | /event_templates.html | 6 built-in templates<br>(Shelter/SAR/HazMat/Weather/Gathering/Exercise) |
| General Info / ICS-201      | /general_info.html    | Initial incident briefing — situation summary, initial actions          |
| ICS Platform — Overview     | /ics/index.html       | Five-section ICS structure navigator                                    |
| ICS — Command Section       | /ics/command.html     | ICS-202 objectives, ICS-208 safety, staff roster, position checklists   |
| ICS — Operations Section    | /ics/operations.html  | T-card board, resource status, assignment, T-card personnel roster      |
| ICS — Planning Section      | /ics/planning.html    | IAP assembly, ICS-205/206/209/211/221 forms, Planning P cycle           |
| ICS — Logistics Section     | /ics/logistics.html   | ICS-205 comms (COML), ICS-206 medical (MEDL), channel library           |
| ICS — Finance/Admin         | /ics/finance.html     | FEMA PA cost tracking, ICS-214 log, expenditure summary                 |
| Planning P Cycle Guide      | /ics/planningp.html   | 15-phase IAP cycle with agenda, required forms, attendees per phase     |
| T-Card Resource Board       | /resources.html       | Add/status/assign resources. T-card personnel roster per card.          |
| NIMS Resource Types         | /resource_types.html  | Pre-loaded NIMS types + custom types. Available in T-card board.        |
| GPS Resource Map            | /resource_map.html    | Live GPS pins per T-card — 3 placement methods, auto-refresh 30s        |
| Starcom Resource Map        | /resmap.html          | Public safety unit positioning map with zone drawing tools              |
| ICS-213 General Message     | /ics213.html          | Formal message with digital signature, print, Winlink export            |
| ICS-214 Activity Log        | /ics214.html          | Unit activity log — export hours directly to FEMA labor costs           |
| ICS-309 Comms Log           | /ics309.html          | Manual communications log with incident archiving                       |
| ICS-204A Briefing Sheet     | /briefing_204a.html   | Assignment briefing for branch/division supervisors                     |
| ICS Form Suite              | /ics-form.html        | Full ICS form set: ICS-202 through ICS-221 with digital signatures      |
| IAP Assembly                | /iap.html             | IAP tracker — all forms for active incident with completion status      |
| IAP Compile (PDF)           | /iap_compile.html     | One-click IAP PDF: title page + section dividers + all forms            |
| FEMA PA Cost Documentation  | /fema_costs.html      | Force Account Labor, Equipment, Materials, Contracts, PW export         |

| PAGE                       | URL                   | NOTES                                                                  |
|----------------------------|-----------------------|------------------------------------------------------------------------|
| FEMA Equipment Rates       | /fema_rates.html      | 44 FEMA equipment categories — edit rates, track rate year             |
| Cost Dashboard             | /cost_dashboard.html  | Live cost totals by category, burn rate, budget tracker, by period     |
| Personnel Accountability   | /accountability.html  | PAR board — ICS-211, T-card personnel, Unaccounted, Cross-Reference    |
| ICS-211 Check-In           | /checkin.html         | Manual ICS-211 check-in form with active list and Check Out buttons    |
| QR/Barcode Scan Check-In   | /scan_checkin.html    | Camera scan or manual ID fallback. BarcodeDetector API, no library.    |
| Member Roster              | /roster.html          | Directory with certs, equipment, QR code generation, CSV import/export |
| Amateur Net Control Logger | /netcontrol.html      | Multi-net, FCC autofill, check-in/out, traffic, ICS-309, observer link |
| Public Safety Net Logger   | /starcom.html         | Radio ID check-in, Starcom/P25 nets, ICS-309 export                    |
| Net Observer Mode          | /observer.html        | Read-only net view, auto-refresh 15s. Share URL with EOC.              |
| Dead Man's Switch          | /deadmans.html        | Audible alert if net goes silent beyond configured interval            |
| Tactical APRS Map          | /tactical.html        | Leaflet — RF APRS always, APRS-IS when WAN up, SARTopo overlay         |
| SARTopo GeoJSON Import     | /sartopo_import.html  | Import planning overlays from SARTopo — persists on tactical map       |
| NEXRAD Radar               | /radar.html           | WAN REQUIRED. Animated NEXRAD loop — play/pause/scrub/speed/palette    |
| NWS/HF Propagation         | /propagation.html     | WAN: SFI, K-index, A-index from NOAA SWPC. Band-by-band conditions.    |
| FCC Callsign Lookup        | /callsign.html        | Offline — 800K+ US licensees in local SQLite. Instant results.         |
| NTS Radiogram Generator    | /nts.html             | ARRL-format traffic with auto word count. Print or log.                |
| Winlink Form Import        | /winlink-import.html  | Import Winlink Express XML forms into incident archive                 |
| AMPRNet Gateway Status     | /amprgate.html        | 44Net WireGuard tunnel status, callsign auth, Part 97 access log       |
| WAN Status Detail          | /wan-status.html      | Both WAN sources, detection method, last check, event log              |
| WAN Settings               | /wan_settings.html    | Configure WAN sources: role, type, detection method, enable/disable    |
| Hospital Directory         | /hospitals.html       | Trauma level, helipad, phone. Feeds ICS-206 medical plan.              |
| Facilities Directory       | /facilities.html      | ICP, staging, shelters, supply depots. Links to T-card assignments.    |
| Repeater Database          | /repeaters.html       | Filter by band/mode/CTCSS. Feeds ICS-205 channel picker.               |
| Channel Library            | /channel_library.html | Pre-loaded interop channels (NIFOG/VTAC). Add custom agency channels.  |
| Grid Square Calculator     | /grid.html            | Grid ↔ lat/lon · distance and bearing between two grids                |

| PAGE                    | URL                       | NOTES                                                                |
|-------------------------|---------------------------|----------------------------------------------------------------------|
| Radio Cheat Sheets      | /cheatsheets.html         | Phonetic alphabet, Q-codes, RST scale, ICS structure, HF frequencies |
| ICS Position Checklists | /position_checklists.html | NIMS activation checklists for all ICS positions                     |
| Meeting Scheduler       | /meetings.html            | Schedule operational briefings — time, location, attendees           |
| Print Center            | /printcenter.html         | Print any ICS form or document to any CUPS printer on EMCOMM-NET     |
| Reference Library       | /refs.html                | Kiwix offline — WikiMed, Wikipedia, iFixit. No internet needed.      |
| Health Monitor          | :5051/health              | Live CPU/memory/disk/temp, all service status, GPS, internet         |
| Pat Winlink Client      | :8090                     | Browser-based Winlink client (Packet, VARA HF, Telnet)               |
| Kiwix Library           | :8080/                    | Offline Wikipedia, WikiMed, iFixit, Wikivoyage                       |

## Service & Port Reference

All services run on the FieldCommand Pi (192.168.50.1) unless noted. The AMPRNet gateway runs on a separate Pi (192.168.50.2).

| PORT | SERVICE           | SYSTEMD UNIT              | DESCRIPTION                                               |
|------|-------------------|---------------------------|-----------------------------------------------------------|
| 80   | nginx             | nginx.service             | Web frontend — all 49 HTML pages, static assets           |
| 5050 | FCC Lookup Server | fieldcommand-fcc.service  | FCC callsign lookup, org config, incidents, net logs      |
| 5051 | ICS Platform      | fieldcommand-main.service | ICS forms, T-cards, check-ins, FEMA costs, GPS, PAR       |
| 8080 | Kiwix Library     | kiwix.service             | Offline reference — WikiMed, Wikipedia, iFixit            |
| 8090 | Pat Winlink       | pat.service               | Winlink email-over-radio — Packet, VARA HF, VARA FM       |
| 631  | CUPS Printer      | cups.service              | USB printer shared to all EMCOMM-NET devices              |
| 2442 | JS8Call           | (laptop app)              | HF digital keyboard messaging — web API on Windows laptop |
| 9000 | AMPRNet Status    | amprgate-status.service   | AMPRNet gateway read-only status — public on EMCOMM-NET   |
| 9001 | AMPRNet Control   | amprgate-status.service   | Tunnel control — localhost only on gateway Pi             |

## Background Services

| SYSTEMD UNIT              | FUNCTION                                                            |
|---------------------------|---------------------------------------------------------------------|
| fieldcommand-main.service | ICS platform server (port 5051) — T-cards, FEMA, PAR, GPS, check-in |
| fieldcommand-fcc.service  | FCC lookup + org config server (port 5050)                          |
| wan-monitor.service       | WAN source monitoring and automatic failover every 30 seconds       |
| wan-monitor.timer         | Timer that keeps wan-monitor running on schedule                    |

| SYSTEMD UNIT                  | FUNCTION                                                         |
|-------------------------------|------------------------------------------------------------------|
| amprgate-status.service       | 44Net gateway status API (ports 9000/9001 on gateway Pi)         |
| amprgate-poll.service         | 44Net WireGuard tunnel keepalive and reconnect on gateway Pi     |
| backup.service / backup.timer | Nightly SQLite backup to labelled USB drive                      |
| kiwix.service                 | Kiwix offline reference library server (port 8080)               |
| pat.service                   | Pat Winlink radio email client (port 8090)                       |
| nginx.service                 | Web frontend — serves all HTML pages and static assets (port 80) |
| cups.service                  | CUPS print server — shares USB printer to EMCOMM-NET             |

## New Feature Configuration

FieldCommand IMS v1.0 includes the following features beyond the base ICS platform. Most require no additional setup — they are available immediately after installation. Items that require initial configuration are noted below.

### Personnel Accountability (PAR)

The Personnel Accountability page (/accountability.html) is available immediately after installation. No additional setup is required. It reads from the same checkin\_entries and tcard\_personnel database tables that are created during normal installation.

| FEATURE                                | SETUP REQUIRED                                                                        |
|----------------------------------------|---------------------------------------------------------------------------------------|
| ICS-211 check-in with Check Out button | None — available at /checkin.html immediately                                         |
| QR/barcode scan check-in               | None — available at /scan_checkin.html. Camera permission required on device.         |
| PAR board with cross-reference         | None — available at /accountability.html immediately                                  |
| T-card personnel roster                | None — Personnel tab appears on every T-card detail panel                             |
| Check-out recording                    | None — Check Out button on every check-in row in checkin.html and accountability.html |

### FEMA PA Cost Documentation

FEMA cost tracking is available immediately. The 2025 FEMA Schedule of Equipment Rates (44 categories) is pre-loaded during installation. Update rates annually at /fema\_rates.html when FEMA publishes a new schedule.

| FEATURE                                     | NOTES                                                               |
|---------------------------------------------|---------------------------------------------------------------------|
| Force Account Labor with fringe calculation | Available immediately. ICS-214 import available from the Labor tab. |
| Equipment with FEMA rate schedule lookup    | 44 rates pre-loaded for 2025. Edit at /fema_rates.html annually.    |
| Materials and Contracts tracking            | Available immediately. Source documentation required for FEMA PW.   |
| Project Worksheet text export               | Click Export PW Text on any incident for a formatted PW summary.    |
| Real-time cost dashboard                    | Available at /cost_dashboard.html. Auto-refreshes every 2 minutes.  |

### Pre-Planned Event Templates

Six built-in templates are pre-loaded during installation: Emergency Shelter Operations, Search and Rescue, Severe Weather Response, Mass Gathering, HazMat Response, and Exercise/Training Scenario. Templates can be edited and new ones created at [/event\\_templates.html](/event_templates.html).

| STEP | ACTION                                                                                                                                          |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Navigate to <a href="http://192.168.50.1/event_templates.html">http://192.168.50.1/event_templates.html</a>                                     |
| 2    | Click the <b>Activate</b> tab — browse the template gallery                                                                                     |
| 3    | Click a template — enter incident name and any custom details                                                                                   |
| 4    | Click <b>Activate Template</b> — a new incident is created pre-loaded with objectives, resource types, channels, and ICS organization structure |
| 5    | To edit a template: click <b>Manage Templates</b> → Edit → modify Objectives, Resources, Channels, or Organization tabs → Save Template         |
| 6    | Export as JSON to share with other FieldCommand systems: click <b>Export JSON</b>                                                               |

## GPS-Tracked Resource Map

The GPS resource map ([/resource\\_map.html](/resource_map.html)) displays color-coded pins for all resources that have a GPS position set. Three placement methods are available.

| METHOD               | HOW TO USE                                                                                                                                                                                                   |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device GPS (browser) | Click Set Position on any resource → click  Use Device GPS. Browser requests location permission. Accuracy shown in meters. |
| Click map to place   | Click  Click Map to Place → crosshair cursor appears → click anywhere on the map to drop the resource pin.                  |
| Manual coordinates   | Type decimal lat/lon directly into the fields. Negative longitude for western hemisphere.                                                                                                                    |

## IAP Export — Print, PDF, and Off-Site Saving

FieldCommand provides three ways to output the IAP, covering both on-site printing and off-site/no-printer scenarios. All three are accessible from the IAP Assembly page ([iap.html](/iap.html)).

| OPTION                                                                                                        | HOW TO USE                                                                                                                                       | BEST FOR                                                                                                          |
|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
|  Print IAP                 | <a href="/iap.html">iap.html</a> → select period and forms → click Print IAP                                                                     | Printer available on EMCOMM-NET or device                                                                         |
|  Save as PDF (recommended) | <a href="/iap.html">iap.html</a> → select period and forms → click Save as PDF. Pi server generates a proper 8.5×11 PDF. Downloads as .pdf file. | Off-site printing, emailing, archiving. Guaranteed correct layout. Copy to USB or email.                          |
|  Save as HTML (fallback)   | <a href="/iap.html">iap.html</a> → select period and forms → click Save as HTML. Browser generates HTML file directly — no server call.          | Fallback if server unreachable. Open in any browser and use File → Print. Page layout may vary by browser.        |
|  Full PDF Compiler         | <a href="/iap_compile.html">iap_compile.html</a> → check forms → click Download IAP PDF (5–15 sec).                                              | Full PDF with cover page, section dividers, embedded signatures. Best for the official distributable IAP package. |

## NEXRAD Animated Radar

The radar page (</radar.html>) requires an active internet connection for radar tile loading. When offline, it displays the last loaded frame with a timestamp. No installation configuration is required — it pulls tiles from NWS RIDGE II directly.

The radar page auto-detects WAN status and resumes loading new tiles automatically when internet connectivity is restored. No manual action is required after a WAN outage.

## Dual-Source WAN Configuration

FieldCommand IMS supports two simultaneous WAN sources with automatic failover. Configure both sources at [/wan\\_settings.html](/wan_settings.html) after installation.

| FIELD            | OPTIONS & DESCRIPTION                                                                                                                                                  |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Role             | Preferred (used first) · Fallback (used when preferred fails)                                                                                                          |
| Type             | Cellular · Satellite · Hotspot · Fixed broadband · Other                                                                                                               |
| Detection method | internet_only — any internet path works (best for hotspots and USB dongles)   ping — checks a specific gateway IP   admin_reachable — checks modem admin interface URL |
| Label            | Human-readable name shown on dashboard: "Cellular", "Satellite", etc.                                                                                                  |
| Enabled          | Toggle a source on or off without deleting configuration                                                                                                               |

  

| STEP | ACTION                                                                                                |
|------|-------------------------------------------------------------------------------------------------------|
| 1    | Navigate to <a href="http://192.168.50.1/wan_settings.html">http://192.168.50.1/wan_settings.html</a> |
| 2    | Click <b>+ Add WAN Source</b> to add a source, or click Edit on an existing one                       |
| 3    | Set Role to <b>Preferred</b> for your primary source (e.g. cellular antenna)                          |
| 4    | Set Role to <b>Fallback</b> for your secondary source (e.g. satellite dish)                           |
| 5    | Choose the Detection Method appropriate for each source type                                          |
| 6    | Click Save. The WAN monitor applies the new configuration within 60 seconds.                          |
| 7    | Use <b>■ Swap Roles</b> to swap preferred and fallback without editing.                               |

## Digital Signature Capture

All 16 Prepared By and Approved By signature fields across the ICS form set support digital signature capture. Signatures are stored as PNG images in the database and are embedded in printed forms and compiled IAP PDFs. No setup required.

To sign: click any Prepared By or Approved By field → signature pad appears → sign with mouse, touchscreen, or stylus → click Accept. Signatures persist through page reloads and appear in the IAP PDF compilation.

## Incident Archive and Beta Reset

| FUNCTION            | HOW TO USE                                                  | NOTES                                                                                                       |
|---------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Archive to USB      | incident_mgmt.html → select incident → Archive to USB       | USB drive must be labelled FIELDCOMMAND. JSON written to <code>/media/fieldcommand/backup/incidents/</code> |
| Restore from USB    | incident_mgmt.html → Restore from USB → select JSON file    | Restores incident to fully active status with all forms, T-cards, costs, and check-ins                      |
| Hard delete from Pi | incident_mgmt.html → select archived incident → Hard Delete | Permanent — confirm archive exists on USB before deleting                                                   |
| Beta/Scenario Reset | incident_mgmt.html → Beta/Scenario Reset → Confirm          | Wipes all incidents, forms, T-cards, costs, check-ins. Preserves roster, channels, repeaters.               |

The USB drive must have volume label FIELDCOMMAND (all capitals). Format as FAT32 or exFAT. Label with: `sudo mlabel -i /dev/sdX1 ::FIELDCOMMAND` (replace sdX1 with actual USB device).

## Maintenance & Updates

```
# Interactive maintenance menu:
sudo bash /opt/fieldcommand/scripts/update.sh

# Menu options:
# 1) Restart all services
# 2) Stop all services
# 3) Check service status
# 4) View live logs
# 5) Refresh FCC database
# 6) Fetch repeater data
# 7) Update web files from current directory
# 8) Backup data to /tmp
# 9) Show disk usage
```

### Loading Repeater Data

The Repeater Database is populated by the Net Control operator or technical lead before each activation. No API token is required for the recommended method.

| STEP | ACTION                                                                                                                                                                                      |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | On any device connected to EMCOMM-NET, open a browser and go to <b><a href="http://192.168.50.1/repeaters.html">http://192.168.50.1/repeaters.html</a></b> .                                |
| 2    | Click the <b>Offline File</b> tab (active by default).                                                                                                                                      |
| 3    | From a device with internet access, log in at <b>repeaterbook.com</b> (free account) → search your county and surrounding mutual-aid area → click <b>Export</b> → <b>CSV</b> .              |
| 4    | Transfer the CSV to a device on EMCOMM-NET (email it to yourself, copy to USB, or download on a device already on EMCOMM-NET).                                                              |
| 5    | Drag the CSV file onto the drop zone, or click to browse and select it.                                                                                                                     |
| 6    | Repeaters load instantly and are saved to the local database — they persist through reboots and are immediately available in the ICS-205 channel picker, channel library, and cheat sheets. |

The CSV export from repeaterbook.com requires only a free account login — it is not the same as the RepeaterBook API token. The API token is a separate application process ([repeaterbook.com/api/token\\_request.php](http://repeaterbook.com/api/token_request.php)) and is only needed if you want the Pi to fetch data automatically via `fetch_repeater.py`. For most deployments, refreshing the CSV before each activation is the practical approach.

### USB Auto-Backup

Insert a USB drive formatted with the label **FIELDCOMMAND** to trigger an automatic backup of all runtime data. The backup copies `/opt/fieldcommand/data/` to a timestamped folder on the USB drive.

```
# Format a USB drive with label FIELDCOMMAND (Linux):
sudo mkfs.vfat -n FIELDCOMMAND /dev/sda1

# Or on Windows: format as FAT32, set volume label to FIELDCOMMAND

# Backup destination on USB:
# /media/fieldcommand/backup/YYYYMMDD_HHMMSS
```

## Troubleshooting

| SYMPTOM                                 | LIKELY CAUSE / FIX                                                                                                                                                                                          |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| http://192.168.50.1 not reachable       | Pi not running, or ASUS router not configured with 192.168.50.x subnet. Check Pi power LED. Verify router LAN IP is 192.168.50.1.                                                                           |
| Dashboard loads but cards give errors   | Not connected to EMCOMM-NET — device is on a different network. Check Wi-Fi — must show EMCOMM-NET.                                                                                                         |
| FCC lookup returns no results           | FCC database not yet built. Run: <code>sudo systemctl start fcc-refresh.service</code> (needs internet)                                                                                                     |
| Repeater database is empty              | No repeater data loaded yet. On any EMCOMM-NET device, open repeaters.html → Offline File tab → download a CSV from repeaterbook.com (free account, no API token needed) → drag the CSV onto the drop zone. |
| APRS map shows no stations              | Direwolf or YAAC not running, or no RF received yet. Check Health Monitor for service status. Confirm antenna connected.                                                                                    |
| Winlink form import fails               | Wrong file type — must be the XML attachment, not the message body. In Winlink Express, right-click .xml attachment → Save As → then import.                                                                |
| Service dot is red on Health Monitor    | Background service has stopped. SSH to Pi: <code>sudo systemctl restart</code>                                                                                                                              |
| Pat Winlink not accessible at port 8090 | pat.service not running. Run: <code>sudo systemctl start pat</code> && <code>sudo systemctl enable pat</code>                                                                                               |
| JS8Call card does not open              | TCP Server API not enabled in JS8Call, or wrong IP entered. Verify: File → Settings → Reporting → Enable TCP Server API, port 2442, hostname 0.0.0.0.                                                       |
| Printer not visible on EMCOMM-NET       | avahi-daemon not running, or CUPS printer not shared. Check: <code>sudo systemctl status avahi-daemon cups</code> . Verify Share This Printer is checked in CUPS UI.                                        |
| Issue: Services crash on startup        | Low disk space or permission error. Check: <code>df -h /</code> and <code>ls -la /opt/fieldcommand/</code> . Verify fieldcommand user owns /opt/fieldcommand/.                                              |

For any issue not covered here — open the Health Monitor at <http://192.168.50.1:5051/health> or check the system log with: `journalctl -u fcc-lookup -n 50`

## Quick Reference Card

| ITEM                      | VALUE                                                                                   |
|---------------------------|-----------------------------------------------------------------------------------------|
| FieldCommand dashboard    | http://192.168.50.1                                                                     |
| Wi-Fi network (SSID)      | EMCOMM-NET                                                                              |
| WAN status dashboard      | http://192.168.50.1/wan-status.html                                                     |
| FieldCommand Pi static IP | 192.168.50.1                                                                            |
| Cellular modem admin      | See your cellular modem documentation for the admin URL (often 192.168.1.1 or 10.0.0.1) |
| WAN data plan             | Manage via your cellular carrier or satellite provider portal                           |
| Satellite dish admin      | See your satellite dish documentation for the admin URL                                 |
| 44Net Gateway Pi IP       | 192.168.50.2                                                                            |
| 44Net gateway status      | http://192.168.50.2:9000                                                                |
| AMPRNet tunnel check      | ping 44.0.0.1 (from any EMMCOMM-NET device)                                             |
| ASUS router admin         | http://192.168.50.254                                                                   |
| CUPS printer admin        | http://192.168.50.1:631                                                                 |
| Pat Winlink (backup)      | http://192.168.50.1:8090                                                                |
| Kiwix offline library     | http://192.168.50.1:8081                                                                |
| Health monitor (raw)      | http://192.168.50.1:5051/health                                                         |
| JS8Call TCP API (laptop)  | http://[laptop-ip]:2442                                                                 |
| Your callsign             | your organization                                                                       |
| your county grid          |                                                                                         |
| Default coordinates       | your deployment coordinates (decimal degrees)                                           |
| Install log               | /var/log/fieldcommand-install.log                                                       |
| Application data          | /opt/fieldcommand/data/                                                                 |
| HTML pages                | /opt/fieldcommand/html/                                                                 |
| Update script             | sudo bash /opt/fieldcommand/scripts/update.sh                                           |